

Worker-Side Discrimination: Beliefs and Preferences

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When agents choose among options with unobserved attributes to maximize expected utility, their choices conflate preferences with beliefs about those attributes. We develop a novel theory-driven identification strategy for such settings, showing that compensating differentials separately identify beliefs from preferences. Noting that workers' preferences and beliefs shape labor market outcomes, yet remain understudied, we apply our method to provide the first evidence on the distribution of workers' preferences over manager gender, and their beliefs about managers' mentoring ability. In the absence of information on manager mentoring ability, workers are indifferent to manager gender. However, upon receiving information on manager mentorship ability, workers prefer to work for female managers, willing to forgo 1.7% of average annual wages. Hence, absent additional information, workers believe female managers are worse mentors (1.6% wage equivalent). Non-parametric estimates of the distributions reveal that 76% prefer female managers with mentoring information, and 67% believe male managers are better mentors absent information. Machine learning algorithms show that individuals with higher education are less likely to hold such beliefs. These beliefs are primarily driven by perceptions that women are less competent. Evidence suggests these beliefs are biased, highlighting scope for information-based policy interventions.

KEYWORDS. Preferences, Beliefs, Identification, Information Experiment, Compensating Differentials, Worker-Side Discrimination, Gender, Mentorship.

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1. INTRODUCTION

Managers differ considerably in their ability to manage and mentor workers, directly affecting workers' careers, wages and within-firm wage inequality, and retention (Fredriksen et al., 2020, Alfonsi et al., 2022, He and le Maire, 2022, Acemoglu et al., 2022, Hoffman and Tadelis, 2021, Lazear et al., 2015). While workers value and sort on many nonpecuniary job benefits (Dey and Flinn, 2005, Blau and Kahn, 2017, Mas and Pallais, 2017, Wiswall and Zafar, 2018, Taber and Vejlín, 2020), and despite substantial public and private investment in mentoring programs,¹ how manager's gender and mentoring quality influence workers' job choices—and whether workers are aware of the mentoring quality of managers—remains an open question. Crucially, observed gender gaps in mentoring outcomes often reflect gender gaps in resources (such as networks) rather than gender gaps in mentoring ability (Schwartz et al., 2022), but worker beliefs about gender gaps in mentorship are unknown.

The job choice of workers, especially jobseekers, depends on—in addition to their preferences—their beliefs (Robinson, 1933, Jäger et al., 2021a) because they may not have complete information about their managers. Information and search frictions contribute to suboptimal labor market matches (Heller and Kessler, 2024, Banerjee and Chiplunkar, 2024) and implicitly assuming workers have correct beliefs can bias policy design and generate welfare losses (Conlon et al., 2018). Understanding the distribution of these preferences and beliefs is essential for designing effective information interventions and understanding gender-based sorting in labor markets, particularly given high training and turnover costs. Additionally, if workers' preferences and beliefs lead them to avoid female managers, they would require wage premiums to accept such positions, thereby strengthening glass ceilings and widening the gender gap at managerial positions in equilibrium.

In this paper, we provide novel evidence on the distributions of worker preferences over manager gender and beliefs about managers' mentoring ability. We focus on mentoring given extensive evidence of its impacts: mentorship enhances human capital accumulation (Falk et al., 2020), wage expectations (Boneva et al., 2021), productivity (Blau et al., 2010), collaborative networks (Ginther and Na, 2021), and promotions (Lyle and Smith, 2014), and helps minorities break glass ceilings (Athey et al., 2000, Müller-Itten and Öry, 2022).² We define worker-side discrimination—in the spirit of a compensating differential (Rosen, 1986)—as a form of selection, where individuals are willing to forgo wages to work for their preferred managers in an otherwise-identical job, extending Becker (1971)'s conceptualization.³ This willingness to trade off wages is driven by pref-

¹For example, the U.S. Department of Justice funds nationwide youth mentoring programs, New York City's Young Men's Initiative provides mentoring at the local level, and public-private partnerships like the Massachusetts Mentoring Partnership coordinate mentoring initiatives statewide. Also see the report on private sector engagement in mentoring by EY and The National Mentoring Partnership (MENTOR) available at: https://www.mentoring.org/wp-content/uploads/2019/12/EY_Full_Report-1.pdf

²The American Economic Association's CeMENT program, where senior women faculty mentor junior women faculty, exemplifies in-group mentoring initiatives.

³Becker (1971) conceptualized worker discrimination in the form of worker disutility from working for a specific group of employers. We extend the concept to incorporate worker beliefs and a tangible measure using compensating wage differentials.

ferences for observable attributes and beliefs about unobservable attributes that workers care about but lack information on.

To identify the distribution of preferences, we design and conduct a hypothetical job choice survey to ensure that demand-side selection, labor market frictions, and omitted variables in general do not confound our results (Blass et al. (2010), Wiswall and Zafar (2018), Ameriks et al. (2020), Fuster et al. (2021), Koşar et al. (2021a), Koşar et al. (2021b)).⁴ Hypothetical choice methods are attractive because they allow unrestricted preference heterogeneity (Blass et al., 2010), enable researchers to hold fixed attributes not considered in the survey through instructions (Wiswall and Zafar, 2018, Koşar et al., 2021a,b) and to document strong correlation between stated and actual choices (Wiswall and Zafar, 2018, Parker and Souleles, 2019, Andrew and Adams, 2025).

To separately identify the distribution of beliefs, we embed a within-worker information experiment where we exogenously vary the observability of managers' mentoring quality. We measure mentorship quality using a five-point rating scale, reflecting firms' growing reliance on manager ratings (Cai and Wang, 2022).⁵

We conduct our hypothetical choice survey and information experiment among job-seeking students at a highly selective university in India one year before graduation. We present respondents with twenty sequential job choice scenarios, each requiring them to choose among three jobs and report wage compensating differentials that would make them indifferent between their preferred job and the alternatives. Jobs are exogenously varied along four realistic attributes: annual wages, flexible hours, manager gender, and manager mentorship quality.⁶ This generates panel data on choices and compensating differentials providing us nonparametric cardinal measures across the support of job attributes.

To circumvent potential social desirability bias in eliciting beliefs about manager mentorship conditional on gender, we implement a within-individual information experiment. In the first ten scenarios ("incomplete scenarios"), respondents observe three job attributes—annual wages, flexible hours, and manager name—with mentorship ratings listed but marked as unavailable. In the final ten scenarios ("complete scenarios"), respondents observe all four attributes including mentorship ratings. Following standard practice, we instruct respondents in every scenario that the jobs do not vary in attributes not mentioned in the survey (Wiswall and Zafar (2018), Koşar et al. (2021b), Koşar et al. (2021a)). Given our novel design, we add a crucial instruction that compensating differentials only increase wages without changing anything else about the job.

⁴Worker heterogeneity in preferences over various dimensions of job attributes, many of which are unobservable to the researcher, is difficult to isolate using data on realized job choices. However, such data have advantages for studying employer discrimination, as employers typically value consistent worker attributes.

⁵Google, e-Bay, and Amazon collect anonymous employee surveys rating managers. Comparably, Completed, TheJobCrowd and Kunukunu are some notable start-ups that provide manager ratings analogous to Glassdoor's firm ratings.

⁶Conceptually, each hypothetical scenario could be thought of as a market. Choice in a market provides individual demand in that market. Survey data on choices over multiple scenarios varying attributes over their support allow us to trace out the individual demand curve. Panel data on choices and compensating differentials over the support of job attributes provide identifying variation for estimation of flexible models.

We use our unique panel data on choices and compensating differentials to estimate a structural model of job choice where worker preferences and valuation of belief parameters can be expressed as willingness to forgo wages. In incomplete scenarios, workers form conditional expectations about unobserved mentorship ratings, so their responses reflect both preferences and beliefs about mentorship conditional on other attributes. In complete scenarios, workers observe all attributes, so responses reflect only preferences. This variation in information availability for each individual identifies individual-specific preferences from within-complete-scenario variation, and between-complete and incomplete-scenario variation identifies individual-specific beliefs, allowing us to identify the full distributions of both parameters.

We find that absent information on manager mentorship, such that choices and compensating differentials are driven by *both preferences and beliefs*, workers are indifferent between male and female managers. However, with information on manager mentorship skill, such that choices and compensating differentials reflect *only preferences*, workers prefer to work for female managers, willing to forgo 1.7% of average annual wages to work for them. This reveals that workers believe female managers are worse mentors. We estimate these negative beliefs to be worth 1.6% of average annual wages, offsetting the preference for female managers in the incomplete scenarios.

Our average estimates mask substantial heterogeneity. Using our within-worker design we estimate individual-specific parameters, and apply nonparametric empirical Bayes shrinkage with log-splines to avoid overstating variance from noisy estimates (Efron, 2016, Walters, 2024). We find that 76% of individuals prefer to work for female managers, while 67% believe female managers are worse mentors than male managers when mentorship information is unavailable. To identify systematic predictors, we use LASSO and find that respondents with advanced degrees (MPhil and PhD) and those whose mothers exceed fathers in educational attainment (specifically fathers with master's and mothers with above-master's degrees) are less likely to hold negative beliefs about female managers' mentoring ability.⁷ LASSO selects no systematic demographic predictors for preferences, suggesting they are driven by idiosyncratic factors rather than observable group characteristics.

To explore potential mechanisms behind the beliefs about gender gap in mentorship, we collected additional data on gender-specific perception of three traits: competence, pleasantness to work with, and non-discriminatory behavior. We extend our model to allow beliefs about mentorship quality to depend on these latent traits allowing us to quantify how much of the gender gap in mentorship beliefs arises through perceived differences in these traits. We find that competence is a significant predictor of mentorship, while the other traits are not. Since males are perceived to be more competent, this channel explains 43.6% of the negative beliefs about female mentorship. Crucially, conditional on these traits, negative beliefs about female mentorship reduce by 62.5%, suggesting that beliefs about gender gaps in mentorship are largely driven by perceived trait differences. Allowing workers to have preferences over these additional

⁷We also find that those with unemployed fathers hold fewer negative beliefs, though this association is approximately one-third the magnitude of both the education effects.

traits shrinks the preference over manager's gender by more than half. This highlights a conceptual strength of our extended framework that empirically engages with concerns discussed in Heckman (1998) on the interpretability of preferences once all dimensions except biological gender are equalized. As we condition on more gender-specific traits, thus making men and women more similar, "pure" gender preferences reassuringly decline.

While we cannot definitively answer whether negative beliefs about female managers' mentorship ability are biased, given the absence of population data on manager mentorship quality in India, understanding potential bias is important for policy design because accurate beliefs require no intervention, while biased beliefs imply welfare losses through suboptimal sorting. An extensive body of evidence from diverse contexts discussed later consistently shows that female mentors are at least as effective as male mentors. More broadly, gender is not a significant predictor of overall managerial effectiveness (Weidmann et al., 2024). This suggests that the negative beliefs we document are likely biased.

We conduct two validation exercises. First, to validate our indirect information experiment approach, we also directly elicit mentorship beliefs by asking respondents to rate managers in hypothetical jobs. Direct elicitation yields opposite results: respondents report female managers as better mentors. This reversal is driven by male respondents. Importantly, these directly elicited beliefs contradict the raw choice data: the share of chosen jobs with female managers rises from 51.8% in incomplete scenarios to 61.1% in complete scenarios, an increase of 18.0%, indicating negative beliefs about female mentorship. This pattern suggests that direct belief elicitation suffers from social desirability bias—consistent with evidence of social desirability bias in the contexts of socially sensitive topics (Bursztyn et al., 2025)—validating our indirect approach to identify beliefs.⁸ Second, we compare preference estimates from our compensating differential approach against those from standard discrete choice methods, since preferences are identified from complete scenarios using either approach. Both methods yield qualitatively similar and economically close estimates, providing a sanity check of our survey design using compensating differentials. Notably, our compensating differential approach yields relatively conservative willingness-to-pay estimates.

Our paper contributes to multiple strands of the literature. Methodologically, we develop a novel identification strategy for information experiments and implement it within the stated-preference literature. While existing methods identify preferences when individuals have complete information,⁹ we advance this literature in three ways. First, we introduce an information experiment that systematically varies the availability of attribute information across scenarios to separately identify preferences and beliefs

⁸While the reversal in directly elicited beliefs could reflect learning from our gender-balanced mentorship ratings, such learning would predict equal assessments of male and female managers, inconsistent with the observed pattern of directly elicited beliefs. Later in the paper, we also discuss different contexts where social desirability bias is less of a concern, and thus direct elicitation methods being more straightforward, should be preferred.

⁹See applications in the contexts of electricity services (Blass et al., 2010), jobs (Wiswall and Zafar, 2018), residential locations (Koşar et al., 2021a), political candidates (Delavande and Manski, 2015), and insurance products (Boyer et al., 2017).

about unobserved attributes. Second, we elicit wage compensating differentials rather than choice probabilities. We formally show that choice probabilities under varying information structures require non-innocuous normalizations that prevent identification, while compensating differentials—being monetary valuations—remain invariant across information conditions, achieving identification. Our framework nests decision problems where a decision-maker (e.g., a worker) chooses among heterogeneous options (e.g., jobs), but does not perfectly observe a subset of the option-specific current state of the world (e.g., manager quality) that is relevant for the decision, and hence must rely on beliefs over these unobserved attributes when comparing expected utilities across options. This setting is pervasive in economic decisions involving unobserved attributes. While information experiments in such settings identify reduced-form effects of information provision (see e.g. [Hastings and Weinstein \(2008\)](#), [Andrabi et al. \(2017\)](#)), they do not identify the underlying belief distribution that any policy intervention would target. Our method fills this gap. By contrast, in concurrent independent work, [Andrew and Adams \(2025\)](#) identify beliefs over *future* states of the world in optimal stopping problems where the *current* state is perfectly observed—a setting fundamentally different from ours. Appendix C discusses how the two frameworks differ and complement each other. Third, our approach provides a method for eliciting beliefs where social desirability bias concerns exist. By using an information experiment to indirectly elicit beliefs, we mitigate the risk of respondents shading their responses toward socially acceptable ones.

Despite extensive evidence on the positive impacts of mentorship on outcomes of mentees across diverse contexts,¹⁰ we estimate for the first time that individuals will forgo up to 5.4% of average annual wages for a 1-SD increase in mentorship rating. The large magnitude is unsurprising given that worker career paths are strongly influenced by their first-time employers ([Arellano-Bover, 2024](#)) and our respondents are first-time jobseekers. The strong preference is consistent with mentoring programs being typically over-subscribed ([Athey and Palikot, 2022](#)). The strong preference for mentorship quality also plays a fundamental role in identification: if workers did not value mentorship, any belief distribution could rationalize the data, leaving beliefs unidentified. It also underscores the substantial welfare cost of biased beliefs.

To the best of our knowledge, we provide the first evidence on the distribution of worker beliefs about the gender gap in mentorship prior to job matching. While extensive research documents that female mentors are at least as effective as male mentors, we show that the majority of workers nonetheless believe female managers are worse mentors, suggesting that these beliefs are likely biased. Thus, while prior literature focuses on gender gaps in mentor effectiveness post-match, we identify the pre-match belief distribution that when biased may prevent optimal mentor-mentee matching. This is particularly consequential given the substantial value workers place on mentorship quality, which we quantify for the first time in the literature, marking our next contribution.

¹⁰See [Athey et al. \(2000\)](#), [Blau et al. \(2010\)](#), [Lyle and Smith \(2014\)](#), [Falk et al. \(2020\)](#), [Müller-Itten and Öry \(2022\)](#), [Germann et al. \(2024\)](#), [Boneva et al. \(2021\)](#)

Our evidence on worker preferences and beliefs conditional on manager gender contributes to the discrimination literature, where worker-side discrimination has received little attention.¹¹ We make three contributions here. First, to the best of our knowledge, we are the first to explicitly allow for worker-side discrimination driven by both beliefs and preferences simultaneously, and to estimate their distributions. The discrimination literature typically estimates average discrimination driven by beliefs (statistical or biased) and preferences (taste-based) *separately*.¹² Second, our design generating unique panel data on compensating differentials allows us not only to test for belief-based discrimination (Altonji and Pierret, 2001, Lange, 2007, Agan and Starr, 2018) but also to quantify it (Bohren et al., 2019) as willingness to forgo wages, going beyond the binary testing approach typical in the literature. Third, while Flory et al. (2015) find that manager gender does not impact application decisions, our results show that this finding is sensitive to the information that workers have about managers and masks substantial heterogeneity.

We also contribute to the literature on social desirability bias. First, consistent with existing evidence, direct elicitation methods are susceptible to social desirability bias.¹³ Second, we find that males are particularly susceptible to social desirability bias when studying beliefs about gender.

Finally, our work contributes to the growing literature on online surveys and experiments (Stantcheva, 2022) that use information treatments to study beliefs¹⁴ by introducing a novel identification strategy and various validation checks.

The paper proceeds as follows. Section 2 describes our hypothetical job choice survey and information experiment. Section 3 presents the sample and data patterns. Section 4 describes the job choice model and Section 5 shows identification using compensating differentials and discusses potential applications. Section 6 discusses estimation and inference, and our main results and heterogeneity analyses. Section 7 examines how gender-specific trait perceptions explain our results. Section 8 discusses whether beliefs are biased followed by policy implications in Section 9. Section 10 presents validation exercises, and Section 11 concludes.

2. INSTITUTIONAL CONTEXT, HYPOTHETICAL JOB CHOICE SURVEY AND INFORMATION EXPERIMENT

We administered our survey to students at a highly selective public university in India who were one year away from graduation. This population allows us to focus on

¹¹Recent work (Abel, 2019, Abel and Buchman, 2020, Ayalew et al., 2021) studies how managers' job performance affects workers' willingness to follow advice—distinct from our focus on mentorship quality given its direct impact on subordinate outcomes (Hoffman and Tadelis, 2021) and the Peter principle (Benson et al., 2019).

¹²See for example Charles and Guryan (2008), Guryan and Charles (2013), Lang and Lehmann (2012), Bertrand and Duflo (2017). Kline et al. (2021) estimate its distribution, but their focus is on firm discrimination toward workers.

¹³See e.g., Bursztyn et al. (2025), Björkman Nyqvist et al. (2018), Kraay and Murrell (2016), Hainmueller and Hangartner (2013), Kuklinski et al. (1997).

¹⁴See Wiswall and Zafar (2015), Kuziemko et al. (2015), Alesina et al. (2019), Boneva and Rauh (2018), Alesina and Stantcheva (2020), Stantcheva (2021), Alesina et al. (2021), Coibion et al. (2022).

high-skill jobseekers. A key institutional feature of campus recruitment in such universities is that candidates typically learn the gender of their prospective manager during final-stage interviews, before accepting job offers.¹⁵ While individuals may switch jobs or teams over time, we intentionally sample those not yet in the labor market to avoid idiosyncratic shocks that may confound their true underlying parameters.

Our hypothetical job choice survey consisted of four parts: (1) instructions to the respondents; (2) twenty hypothetical job choice and compensating differential *scenarios*, within which the information experiment was embedded; (3) direct belief elicitation; and (4) demographic questions. The structure is schematically represented in Figure 1.¹⁶ We describe each component in detail below.

2.1 Instructions

The first part of the survey presented definitions of the exogenously varied job attributes—manager name, annual wages, flexible hours, and manager rating—as shown verbatim in Figure G.1a. Manager rating was defined as: “... *the average rating of the mentorship of the manager, provided by this manager’s current employees in an anonymous survey. This is a measure of how good of a mentor this manager is to their subordinates.*” followed by the description of the numeric five-point scale. Such anonymous employee surveys are standard practice in large firms such as Amazon, Google, and eBay, and were similarly used in the field experiment of Cai and Wang (2022).

We then presented respondents with instructions that support our identification strategy (see Appendix Figure G.1b; Wiswall and Zafar (2018), Koşar et al. (2021b)). Two key instructions were emphasized. First, jobs did not differ along any attribute “...*NOT MENTIONED...*” in the survey. Second, reported wage increases should be interpreted as affecting only pay, with all else about the job held constant. These instructions were reiterated within each scenario. Respondents were next shown two example scenarios before beginning the main survey.

2.2 Job Choice Scenarios with Compensating Differentials

Each respondent completed twenty hypothetical job choice scenarios. In each, they selected one of three jobs and then reported the minimum wage increase required to make them indifferent between their chosen job and each of the two unchosen jobs—referred to as compensating differentials. The first 10 scenarios formed the “incomplete” set, in which manager mentorship ratings were mentioned but not shown. The final 10 were the “complete” scenarios, where ratings were provided. This within-subject variation forms the core of our information experiment. Respondents entered compensating differentials using a slider ranging from 0 to 2 lakh INR ($\approx$ \$9,492 in PPP). If their required

¹⁵See Alam et al. (2026b) Online Appendix A.1 for further institutional details on campus recruitments and Online Appendix A.2 for details on the administration and implementation of the survey.

¹⁶The survey design, data and code required to reproduce the analysis are provided in the replication package (Alam et al., 2026a).

compensation exceeded 2 lakh, they were prompted to enter a higher value on a subsequent screen.¹⁷ Tables 1 and 2 show representative examples from the incomplete and complete sets, respectively. For U.S.-adapted examples see Appendix Tables G.1 and G.2.

Table 3 presents descriptive statistics of the job attributes (Panel A), balance across manager gender (Panel B), and balance across complete and incomplete scenarios (Panel C). Across the 20 scenarios, respondents saw 60 unique jobs, evenly split between male and female managers and balanced across complete and incomplete scenarios. While variation in job attributes was exogenous, it was not fully random. Scenarios were designed to avoid dominant options (Wiswall and Zafar, 2018). Approximately half the jobs offered flexible hours. Average annual wages were 7.11 lakhs INR ($\approx$ \$33,745 in PPP), consistent with offers to past graduating cohorts at the sampled university. The variation in wages was deliberately small, reducing concerns that some jobs might be interpreted as entry- vs. senior-level. The average mentorship rating in complete scenarios was 3.41.¹⁸

2.3 Remaining Parts of the Survey

2.3.0.1 Direct Belief Elicitation: Following the 20 job choice scenarios, we directly elicited respondents' beliefs about manager mentorship ability. This exercise allows us to compare beliefs inferred from the information experiment with those elicited directly. Respondents were shown 10 jobs—each with a manager's name, wage, and flexibility status—and asked to report each manager's mentorship ratings on a 0–5 sliding scale that they expected (see Appendix Figure G.3).

2.3.0.2 Demographic Questions: The final section of the survey collected demographic information on respondents' field of study (arts, science, or engineering), family income, and parental education and occupation. We also included questions to assess whether respondents followed the survey instructions. The survey concluded with a prompt to select a preferred mode of online payment and its associated details for compensation of INR 500 ($\approx$ \$24 in PPP).¹⁹

2.4 Key Highlights of our Design

This subsection outlines key elements of our hypothetical job choice survey and the embedded information experiment that enable identification of beliefs and preferences from reported choices and compensating differentials.

In Section 5.4, we discuss why the presence of incomplete scenarios to identify beliefs requires compensating differentials rather than choice probabilities. This results from choice probabilities requiring non-innocuous normalizations. See Appendix B for a formal proof.

¹⁷We did not find any evidence of any design-induced bunching of reported compensating differentials at the boundary of the slider. We thank Jeff Smith for bringing our attention to check this.

¹⁸We return to the informativeness of the rating in the results section, where we address concerns of other interpretations that could have stemmed from not providing more context about the jobs.

¹⁹The purchasing power parity of 1 USD in 2019 is equivalent to 21.07 INR. Source: <https://data.oecd.org/conversion/purchasing-power-parities-ppp.htm>

To signal and vary gender, we used only managers' first names. In the Indian context, last names can reveal other social identities like caste and/or religion. By omitting them, we avoided unintended variation across social categories. All names used were common and unambiguously male or female, ensuring that gender was the only varying dimension. This approach avoids potential confounds that arise in other contexts, such as the U.S., where first names may correlate with both gender and race (see e.g., [Bertrand and Mullainathan \(2004\)](#), [Kline et al. \(2021\)](#)).

A key advantage of our design is that we observe the full choice set by construction. For each scenario, we know which job was chosen. The compensating differentials for the unchosen jobs provide a nonparametric cardinal measure of utilities and allows us to avoid making any distributional assumptions on the preference or belief parameters. The data on the compensating differentials also allow us to directly estimate and interpret the parameters as measures of willingness to pay or to forgo wages.²⁰

The information treatment is given to every individual. For each individual, we observe the sequence of choices made and the compensating differentials reported over the incomplete scenarios and then over the complete scenarios. This within-subject design allows us to recover the distributions of preferences and beliefs and not just the first moment, which would have been the case had we used a between-subject design.

In the incomplete scenarios, to avoid priming respondents to assume either similarity or systematic differences in manager ratings, we used the following wording: *"The rating of each manager may be different, but the data are not available."*

A key advantage comes from our indirect approach to recover beliefs as direct elicitation may be susceptible to social desirability bias. To assess this, we compare our indirect approach with those from directly elicited beliefs after the information experiment. As we show later in Section 10.1, estimated model-based beliefs using our indirect approach differ sharply from directly elicited beliefs—particularly among male respondents—validating the usefulness of our indirect method in settings where social desirability bias is of concern.

We use 20 scenarios to feasibly span and vary over the support of job attributes while managing cognitive load.²¹ Asking respondents to choose among many jobs per scenario, and over a larger number of scenarios proved infeasible during piloting. Instead, we asked them to evaluate three options in each scenario across multiple scenarios. This generated rich panel data on choices and compensating differentials across jobs that exogenously vary in attributes.

3. DATA

3.1 Sample Description

Eligible participants were students no more than one year from graduation. A total of 626 individuals participated in the survey, and sample selection based on verification and

²⁰This can also be achieved with data on choice probabilities with an additional step to transform the estimates into willingness-to-pay measures.

²¹Similar surveys use 8–24 scenarios (e.g., [Wiswall and Zafar \(2018\)](#); [Koşar et al. \(2021a\)](#)).

non-missingness of the data used gives us a final sample of 580 survey respondents.²² Among these respondents, 41.72% identified as female.²³ The median completion time was 51.4 minutes.

Table 4 reports descriptive statistics. Among respondents, 44.7% were enrolled in arts and humanities, 33.3% in engineering, and 22.1% in science. Female students were predominantly in arts and humanities (71.9%), while male students were predominantly in engineering (49.4%) and science (25.4%). We also observe gender differences in family background. On average, female respondents come from more socioeconomically advantaged households. This pattern is consistent with the evidence that gender gap in tertiary education is substantially lower in rich households than in poor households in India (Choudhury and Kumar, 2024). Females from relatively more privileged households are less likely to face gender-norm-based social frictions, financial frictions, and have lower opportunity costs of time to attend colleges and universities (Sekhri et al., 2024). Our sample characteristics in terms of family income and parental education are comparable to studies using data from other Indian colleges such as Dasgupta et al. (2022) who show their sample representativeness with the associated university population. The selective nature of the elite university of our sample reflects some differences when compared with the population of college and university students in India. The proportion of females in Indian universities in the 2021-22 academic year was 5.3pp higher at 47% (Ministry of Education, Government of India, 2023). The national average enrollment rate in Engineering was 11.8% and in Science was 14.8%. The proportions of our survey respondents majoring in Engineering, Science and Arts follow overall and gender-specific national patterns, although our sample is relatively over-represented in Engineering and Science, while being relatively under-represented in Arts and humanities, due to specialization.

3.2 Patterns in the Raw Data

We begin by examining how individuals' choices and reported compensating differentials vary across the complete and incomplete scenarios by manager gender. These descriptive patterns motivate the job choice model introduced in the next section.

Panel A of Table 5 reports the percentage of jobs chosen by manager gender. Choice patterns are similar across male and female respondents. In the incomplete scenarios—where mentorship information is withheld—jobs with male and female managers are chosen at similar rates. In contrast, in the complete scenarios, where mentorship

²²We dropped 13 respondents (7 whom we could not trace back to our email distribution, and 6 more whom we could not cross-verify as students). All respondents who completed more than 85% of the survey were paid. Applying this completion threshold dropped 33 more respondents. Of these 33 respondents, 26 did not complete the main choice and compensating differentials block and, by construction, could not complete the further blocks of questions on direct elicitation, traits, and demographics; the remaining 7 did not complete the rest of the survey. This sample selection ensures no missing values in the variables used for all estimations.

²³We asked respondents to report their biological sex. We did not collect data on gender identities that differ from biological sex.

ratings are shown, 61.1% of chosen jobs have female managers, 22.3 percentage points more than male-managed jobs.

Panel B reports average compensating differentials for unchosen jobs by manager gender. In the incomplete scenarios, respondents demand 6.3 thousand INR ($\approx$ \$300) more to choose jobs with female managers over male managers. This pattern reverses in the complete scenarios: respondents now require 6.1 thousand INR ($\approx$ \$290) more to switch from a female-managed to a male-managed job. Both differences are statistically significant at the 1% level. While these unconditional comparisons involve only unchosen jobs, they are informative. A fuller interpretation requires modeling both the extensive margin (choice) and the intensive margin (compensating differential) as functions of job attributes. This is precisely what our job choice model does. Finally, before turning to the model, we address a natural question in this setting: are female respondents more likely to choose jobs with female managers?

3.3 Testing for In-Group Preferences

We next ask whether female respondents are more likely than male respondents to choose jobs with female managers. We test this using a difference-in-differences strategy applied to data from the complete scenarios.

Let $i = 1, \dots, N$ index individuals, $j = 1, \dots, J$ index jobs, and s index scenarios. Define $Choice_{ijs}$ as an indicator equal to 1 if individual i chooses job j in scenario s , and 0 otherwise. We estimate the following model:

$$\begin{aligned}
 Choice_{ijs} = & \delta_0 + \delta_1 \mathbb{I}(g_i = f) \mathbb{I}(MG_{j(s)} = f) + \delta_2 \mathbb{I}(g_i = f) + \delta_3 \mathbb{I}(MG_{j(s)} = f) \\
 & + Attributes'_{j(s)} \gamma_1 + Demographics'_i \gamma_2 + \lambda_s + e_{ijs}
 \end{aligned} \tag{1}$$

Here, g_i is respondent i 's gender, and $MG_{j(s)}$ is the gender of the manager in job j of scenario s . The vector $Attributes_{j(s)}$ includes all job attributes other than manager gender—annual wages, flexible hours, and mentorship rating. $Demographics_i$ includes the respondent's background characteristics. We include scenario fixed effects λ_s to leverage the variation in choices made within scenarios resulting from the variation in attributes between jobs within each scenario. We estimate this model using a logit specification and bootstrap standard errors at the individual level.

Table 6 presents the marginal effect estimates from equation (1). We find no evidence that female respondents are more likely to choose jobs with female managers. This is consistent with the raw data, which also show no systematic gender differences in choices or compensating differentials. As expected, respondents are more likely to choose jobs with higher wages, flexible hours, and higher mentorship ratings.

We now turn to the structural model of job choice, which allows us to formally incorporate compensating differentials and isolate the role of beliefs in the absence of information about manager mentorship from the role of preferences with parameters interpreted in monetary terms as percentages of average annual wages.

4. MODEL

Individuals are indexed by $i \in \{1, \dots, N\}$, and jobs are indexed by $j \in \{1, \dots, J\}$. Let X_j denote a K -dimensional vector of attributes of job j over which individuals have preferences. The utility of an individual i from job j is given by

$$U_{ij} = u_i(X_j) + \epsilon_{ij} \quad (2)$$

where ϵ_{ij} denotes all unobservables that affect the utility of individual i from job j . Individuals use expected utilities while reporting their job choice and the corresponding compensating differentials.

Individuals have preferences over working for a male manager (G), annual wages (W), availability of flexible hours (H) and manager mentorship rating (R). We denote this set of attributes as $X \equiv \{G, W, H, R\}$. In the complete scenarios, respondents observe X for each job. In the incomplete scenario, respondents observe $\tilde{X}$, where $\tilde{X} \equiv X \setminus R$. In the incomplete scenarios, when individuals do not observe the mentorship rating R , they use their beliefs about R given $\tilde{X}$ to form their expected utilities.

The model is nonparametrically identified up to the distribution of $\epsilon_i \equiv \{\epsilon_{i1}, \dots, \epsilon_{iJ}\}$, as shown in Appendix A. In the following sections, to keep things simple, we use a linearly separable model. The utility of an individual i with preference parameter vector $\beta_i \in \mathbb{R}^K$ from job j with K dimensions of attributes X_j is given by

$$U_{ij} = X_j' \beta_i + \epsilon_{ij} \quad (3)$$

Identification of more variants of the model allowing for various interactions is shown in Appendix F and monotone transformations is shown in Appendix F4.

4.1 Complete Scenarios

In the complete scenarios, individuals observe all attributes in set X for each job. The expected utility of individual i from job j conditional on its observable attributes in the complete scenarios is given by

$$\mathbb{E}_i[U_{ij} | X_j] = X_j' \beta_i + \mathbb{E}_i(\epsilon_{ij} | X_j) \quad (4)$$

The preference parameters of individual i is given by the vector $\beta_i \equiv (\beta_i^G, \beta_i^W, \beta_i^H, \beta_i^R)'$. We assume that each individual i knows their preferences β_i^x for each attribute $x \in X \equiv \{G, W, H, R\}$ and hence do not take expectations over them.

4.2 Incomplete Scenarios

Denote the set of observable attributes in job j as $\tilde{X}_j \equiv X_j \setminus \{R_j\}$ in the incomplete scenarios. Individual i uses their beliefs $R_j = \tilde{X}_j' \alpha_i^x + \eta_j$ where η_j could be interpreted as measurement error. Consequently, individual i forms expectations on the mentorship of the manager in the associated job as

$$\mathbb{E}_i(R_j | \tilde{X}_j) = \tilde{X}_j' \alpha_i \quad (5)$$

The belief parameters of individual i are given by the vector $\alpha_i \equiv (\alpha_i^G, \alpha_i^H, \alpha_i^W)'$. We assume that all individuals know their belief parameters α_i^x for each attribute $x \in \tilde{X} \equiv X \setminus \{R\}$ and hence do not take expectations over them. Observe that $\alpha_i^G = \mathbb{E}_i(R \mid G = \text{male}, W, H) - \mathbb{E}_i(R \mid G = \text{female}, W, H)$ represents how much on average individual i believes a male manager's mentorship rating differs from that of a female manager.

It is important to emphasize that the expectations here are allowed to vary by individuals. This allows individuals to draw from different distributions of mentorship, which may not necessarily be the true distribution. Also note that linear separability of the belief function is a simplifying assumption which does not aid in identification. Indeed, we could allow for and identify parameters on various interactions among the attributes $\tilde{X}_j \equiv X_j \setminus \{R_j\}$ observable in the incomplete scenarios. As the number of scenarios in both the complete and incomplete scenarios approach infinity we could allow for a fully non-parametric belief function.

The expected utility of individual i from job j conditional on its observable attributes $\tilde{X}_j$ in the incomplete scenarios is given by

$$\begin{aligned} \mathbb{E}_i[U_{ij} \mid \tilde{X}_j] &= \sum_{x \in \tilde{X}} \beta_i^x x_j + \beta_i^R \mathbb{E}_i(R_j \mid \tilde{X}_j) + \mathbb{E}_i(\epsilon_{ij} \mid \tilde{X}_j) \\ &= \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) x_j + \mathbb{E}_i(\epsilon_{ij} \mid \tilde{X}_j) \end{aligned} \quad (6)$$

Denote for each attribute $x \in \tilde{X} \equiv \{G, W, H\}$ and each individual i :

$$\tilde{\beta}_i^x \equiv \beta_i^x + \beta_i^R \alpha_i^x \quad (7)$$

Observe that $\tilde{\beta}_i^x$ is comprised of two terms: the preference parameter β_i^x for attribute x and how much x affects the belief about the manager's rating α_i^x , weighted by how much the individual cares about the manager's rating β_i^R .

5. IDENTIFICATION

In this section, we show how our experimental panel data on choices and compensating differentials identify the preference and belief parameters of our model by exploiting variation in the reported compensating differentials within and between the complete and incomplete scenarios.²⁴ Through survey instructions (presented verbatim in Appendix Figure G.1b), individuals were instructed to assume that:

Assumption (1): All attributes not mentioned in the survey were the same for all jobs.

Assumption (2): The reported compensating differential would increase only wages and change nothing else about the job.

Observe that instruction 1 is an assumption between jobs, while instruction 2 applies within jobs. The purpose of these instructions was to ensure that there was no selection on attributes not mentioned in the survey. [Wiswall and Zafar \(2018\)](#) delineate the

²⁴In the Appendix, we also specify a more flexible model where the rating variable is used as a signal for overall manager quality and show identification of beliefs in that setting.

importance of assumption (1) in a set-up such as our own in contrast to the settings in audit studies on hiring discrimination, where there is little preventing employers from making different assumptions about different job applicants conditional on the observables in their resumes.²⁵ Additionally, [Wiswall and Zafar \(2018\)](#) explain that assumption (1) via the instructions avoids biases that could arise from omitted variables through unobservables or from potential equilibrium effects in realized choice data. For both scenarios, assumption (2) implies that the compensating differential increases only the wage and does not change the conditional expectation of the unobservables. Note that for the incomplete scenarios, it applies to the conditional expectation of the manager rating as well, as we show in the following sections. We use the data on compensating differentials to equate the expected utilities in the complete and incomplete scenarios. These two assumptions, which form a clear parallel to the instructions given to the respondents, form the basis of our identification strategy. See [Appendix A](#) for a proof of nonparametric identification of the model. Here we show identification in the parametric model.

5.1 Preference Parameters

We show the identification of the preference parameters β_i^x for all $x \in \{G, H, R\}$ leveraging variation in compensating differentials within the complete scenarios.

Implication of Assumption (1): The instructions imply that unobservables across different jobs in conditional expectations are the same within each scenario. For every individual i and every job $j \neq k$ within each complete scenario,

$$\mathbb{E}_i(\epsilon_{ij} \mid X_j) = \mathbb{E}_i(\epsilon_{ik} \mid X_k)$$

Implication of Assumption (2): In the complete scenarios, for each job k , individuals observe the vector of attributes $X_k \equiv \{G_k, H_k, W_k, R_k\}$. Suppose that individual i chooses job j and then provides a compensating differential of Δ_{ijk} that she would require to choose job k instead. The instructions imply that for unchosen jobs such as job k ,

$$\mathbb{E}_i(\epsilon_{ik} \mid X_k, \Delta_{ijk}) = \mathbb{E}_i(\epsilon_{ik} \mid X_k) \quad (8)$$

Given the above, equating the expected utilities between job j and job k with the provided compensating differential of Δ_{ijk} and normalizing $\beta_i^W = 1$, we have

$$\begin{aligned} \mathbb{E}_i(U_{ij} \mid X_j) &= \mathbb{E}_i(U_{ik} \mid X_k, \Delta_{ijk}) \\ \Delta_{ijk} &= (X_j - X_k)' \beta_i \end{aligned} \quad (9)$$

Since $(X_j - X_k)$ is of full rank stacking all j, k ensured by our survey, the preference parameters $\beta_i \equiv \{\beta_i^G, \beta_i^H, \beta_i^R\}$ are identified from the variation in the reported compensating differentials in the complete scenarios under assumptions (1) and (2).

²⁵See [Heckman \(1998\)](#) for more discussion on the use of audit studies in detecting discrimination.

5.2 Belief Parameters

We show identification of the belief parameter vector $\alpha_i \equiv (\alpha_i^G, \alpha_i^H, \alpha_i^W)'$ for each individual i , leveraging variation in the reported compensating differentials between the complete and incomplete scenarios.

Implication of Assumption (1): The instruction implies that unobservables affecting utilities and beliefs across different jobs in conditional expectations are the same within each scenario. For every individual i and every job $j \neq k$ within each incomplete scenario,

$$\mathbb{E}_i(\epsilon_{ij} \mid \tilde{X}_j) = \mathbb{E}_i(\epsilon_{ik} \mid \tilde{X}_k)$$

Implication of Assumption (2): In the incomplete scenarios, for each job k , individuals observe the vector of attributes $\tilde{X}_k = \{G_k, H_k, W_k\}$. Suppose that individual i chooses job j and provides a compensating differential of $\tilde{\Delta}_{ijk}$ that she would require to choose job k instead. All the compensating differential does is increase the wages in job k by $\tilde{\Delta}_{ijk}$. The implication of assumption (2) is that it has no effect on the conditional expectation of managers' ratings or on the conditional expectation of the unobservables affecting utility. That is,

$$\mathbb{E}_i(R_k \mid \tilde{X}_k, \tilde{\Delta}_{ijk}) = \mathbb{E}_i(R_k \mid \tilde{X}_k) \quad (10)$$

Thus, the expected utility from job k taking into account the compensating differential of $\tilde{\Delta}_{ijk}$ along with the beliefs about mentorship in equation (5) and normalizing β_i^W to 1 is:

$$\begin{aligned} \mathbb{E}_i(U_{ik} \mid \tilde{X}_k, \tilde{\Delta}_{ijk}) &= \beta_i^G G_k + \beta_i^H H_k + (W_k + \tilde{\Delta}_{ijk}) + \beta_i^R \mathbb{E}_i(R_k \mid \tilde{X}_k, \tilde{\Delta}_{ijk}) + \mathbb{E}_i(\epsilon_{ik} \mid \tilde{X}_k, \tilde{\Delta}_{ijk}) \\ &= \beta_i^G G_k + \beta_i^H H_k + (W_k + \tilde{\Delta}_{ijk}) + \beta_i^R \mathbb{E}_i(R_k \mid \tilde{X}_k) + \mathbb{E}_i(\epsilon_{ik} \mid \tilde{X}_k) \\ &= \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) x_k + \tilde{\Delta}_{ijk} + \mathbb{E}_i(\epsilon_{ik} \mid \tilde{X}_k) \end{aligned}$$

We can normalize $\beta_i^W = 1$ as discussed before, because the valuation of a dollar remains a dollar irrespective of whether the scenario is complete or incomplete. Equating the expected utilities between job j and k conditional on the compensating differential $\tilde{\Delta}_{ijk}$ under A1, we have

$$\begin{aligned} \mathbb{E}_i(U_{ij} \mid \tilde{X}_j) &= \mathbb{E}_i(U_{ik} \mid \tilde{X}_k, \tilde{\Delta}_{ijk}) \\ \tilde{\Delta}_{ijk} &= \sum_{x \in \tilde{X}} \underbrace{(\beta_i^x + \beta_i^R \alpha_i^x)}_{\equiv \tilde{\beta}_i^x} (x_j - x_k) \end{aligned} \quad (11)$$

Thus, once β_i^x and $\tilde{\beta}_i^x$ are identified, α_i^x is identified $\forall x \in \tilde{X}$ as long as $\beta_i^R \neq 0$ and $\tilde{X}_j - \tilde{X}_k$ is of full rank stacking all j, k .

$$\alpha_i^x = \frac{\tilde{\beta}_i^x - \beta_i^x}{\beta_i^R} \quad (12)$$

In the complete scenarios, the compensating differentials are a function of how jobs j and k vary in their attributes, weighted by how much individual i cares about each of those attributes (equation (9)). In contrast, in the incomplete scenarios (equation (11)), they are a function of how jobs j and k vary in their attributes apart from mentorship, weighted by not only how much individuals care about each attribute but also by how much they believe each attribute is correlated with mentorship skill—very much in the spirit of omitted variable bias.

There are two circumstances under which beliefs α_i^x are not identified. The first is when i does not care about mentorship ability i.e., $\beta_i^R = 0$. This makes the variation in the observed choices and compensating differentials independent of mentorship skill. Consequently, any belief distribution can rationalize the observed data. The second is when individuals believe mentorship is independent of all observed attributes with separable preferences. In that case, for each individual i , $\mathbb{E}_i(R_k | \tilde{X}_k) = \mathbb{E}_i(R_k)$ for all jobs k , which is a constant, though it could vary by i . Under additively separable utilities, this constant enters expected utility identically across all jobs and therefore cancels in within-individual compensating differentials, making beliefs unidentified. Identification of some belief moments under independence would require non-separable preferences (see Appendix A).

5.3 Applications

Our method applies where agents compare multiple options each with its own draw of an unobserved attribute, such that the current state of the world is not fully observed.²⁶ Examples of applications include school choice with unobserved school quality, physician choice with unobserved provider quality, housing choice with unobserved neighborhood quality, firm hiring of workers with unobserved worker productivity, etc. Respectively, researchers could study parental beliefs about school quality by varying school performance metrics and eliciting tuition compensating differentials, beliefs about physician quality by varying provider quality information and eliciting fee differentials, beliefs about neighborhood quality by varying neighborhood information and eliciting willingness to pay for housing/rent differentials, and employer beliefs about worker quality by varying performance information and eliciting wage offers.

5.4 Non-identification with Choice Probability Data

Before proceeding to estimation, we address why identification—and consequently our survey design—relies on compensating differentials rather than choice probabilities or ranks commonly used in stated preference literature. Choice probability data require assumptions about idiosyncratic taste shocks (typically Type I extreme value) and an associated variance normalization—standard in discrete choice models when all scenarios provide complete information.

²⁶For identification of beliefs in optimal stopping problems where the current state of the world is fully observed, and the only uncertainty is about the future states of the world see [Andrew and Adams \(2025\)](#). For a comparison of our method with the method in [Andrew and Adams \(2025\)](#), see Appendix C.

However, any design such as ours that requires variation of information structures changes what remains unresolved at the time of choice. In incomplete scenarios, mentorship quality is unresolved, so choice probabilities must integrate over this additional source of uncertainty absent in complete scenarios. The total variance of unobserved components entering choice probabilities thus differs systematically between complete and incomplete scenarios. Choice probabilities in incomplete scenarios contain variance arising both from idiosyncratic taste shocks and belief-formation, while complete scenarios contain only the former.

Crucially, the individual faces this unresolved uncertainty regardless of what the researcher elicits. The difference lies in what the researcher must do to recover structural parameters from the elicited object. To recover parameters from choice probabilities, the researcher must normalize variance in both scenario types. Since the underlying variance structures differ, these normalizations become non-innocuous. Thus, one cannot distinguish whether differences in choice probabilities reflect belief parameters or normalization artifacts. Appendix B provides the formal proof. This non-identification result persists even with parametric assumptions (e.g., zero mean normal distributions for both components; see Appendix B.1).²⁷

Compensating differentials achieve identification because monetary valuations remain invariant across information structures. A dollar forgone represents the same expected utility loss regardless of information availability, enabling direct comparison across scenarios without confounding scenario-specific variance normalizations.

6. ESTIMATION AND MAIN RESULTS

Compensating differentials are reported with two independent classical measurement errors that enable joint estimation of the individual indifference conditions in the complete and incomplete scenarios in equations (9) and (11) respectively.²⁸ Normalizing $\beta_i^W = 1$, we block bootstrap at the respondent level to allow for arbitrary within-respondent correlation. See Appendix D for full estimation details.

6.1 *Model Estimates of Preference and Belief Parameters*

We present estimates of preference and belief parameters from our job choice model. We first present estimates on the demand for mentorship quality, followed by preferences for manager gender, and finally beliefs about the gender gap in mentorship. This ordering builds the logical foundation: understanding how much workers value mentorship is necessary for identification and for interpreting both gender preferences in the presence of information on mentorship and beliefs about gender differences in mentorship ability in the absence of its information.

²⁷Using rank data is always dominated—even in complete scenarios—because cardinal information provides richer information on the distribution of preferences to the researcher than ordinal information from ranks. Hence, non-identification with rank data follows similar steps after specifying a distribution of the unobserved part of the utility function.

²⁸The two classical measurement errors are rounding to multiples of five and slider interface noise. The former is also documented to be common in surveys asking for choice probabilities (Blass et al., 2010). See Figure 2. Being classical in nature, they only affect standard errors but not consistency.

Since the utility parameter on wages is normalized to 1 and wages are in units of hundred thousand INR, the estimates should be interpreted as valuations of each attribute in units of hundred thousand INR. For better interpretability, we convert these to percentages of average annual wages.

6.1.1 Demand for Mentorship Quality Workers highly value mentorship quality. Table 7 Panel A shows workers will forgo 11.1% of average annual wages ($\approx$ \$3,800 or 80 thousand INR) for a one-point increase in mentorship rating on our five-point scale. In standard deviation units, workers value a one-SD increase in mentorship at 5.4% of annual wages. Under diminishing returns to mentorship, a marginal increase in mentorship ability is potentially of higher value to first-time jobseekers than to experienced workers.

A potential concern is whether the mentorship rating was informative to respondents. Recall that the mentorship rating was presented in the survey as "*... the average rating of the mentorship of the manager, provided by this manager's current employees in an anonymous survey.*" Although our scenarios do not specify job types or industries to reduce cognitive load, if respondents were thinking of jobs dominated by out-group workers or different job types that made the mentorship ratings uninformative, we would observe evidence of this in the data. By contrast, the lower limit of the 95% confidence interval is 10.3% of average annual wages, far from zero. The detailed instructions and this evidence mitigate concerns that the mentorship ratings were uninformative to respondents.²⁹

6.1.2 Preference for Working for Female Managers Table 7 Panel A presents estimates from jointly estimating equation system (D.1). In complete scenarios, where all job attributes are observed, we identify pure preference parameters. The first key finding from the complete scenarios is a strong preference for female managers. On average, workers are willing to forgo approximately 12 thousand INR ($\approx$ \$570) or 1.7% of average annual wages to work for a female rather than male manager ($\beta^G = -0.119$). The 95% confidence interval ranges from 1.2% to 2.2% of average annual wages.

In incomplete scenarios, the "biased" estimate of the preference for male managers ($\tilde{\beta}^G$) is small and statistically indistinguishable from zero. This stark difference between complete (β^G) and incomplete scenarios ($\tilde{\beta}^G$) reveals that workers' indifference masks preferences for female managers being offset by negative beliefs about their mentorship ability. This is evidence of belief-based discrimination only if workers value mentorship quality—which we discuss next.

6.1.3 Beliefs about the Gender Gap in Mentorship The estimates of the belief parameter on male managers' mentorship (α_i^G) are obtained from the estimates of the vectors ($\tilde{\beta}_i^G, \beta_i^G, \beta_i^R$) using equation (12). Table 7 Panel B column (1) shows that on average workers believe male managers have 0.14-point higher mentorship ratings than female

²⁹There is less concern about alternative interpretations of mentorship. For example, it is unlikely that English majors would answer our questions by imposing on themselves the extra cognitive load of thinking about jobs outside their domain of specialization.

managers (0.29 SD). The estimates of α_i^G are hard to interpret since they represent relative beliefs about a five-point scale. A more interpretable measure of beliefs—in monetary terms—is the valuation of beliefs about male manager mentorship ($\beta_i^R \alpha_i^G$). This is obtained by the difference in the parameters between the complete and incomplete scenarios: $\tilde{\beta}_i^x - \beta_i^x = \beta_i^R \alpha_i^x$ for all $x \in \{G, W, H\}$. In Table 7 Panel B column (2), we report that workers' negative beliefs about female mentorship are worth 1.6% of average annual wages ($\beta_i^R \alpha_i^G$).³⁰

Note that classical measurement error in the reported compensating differentials leading to inflation of standard errors in the estimation of the preference parameters will trickle down to the standard errors of the belief parameters. Despite this, the belief parameter estimate is statistically significant. The estimates in conjunction with the model and the evidence that mentorship skill is a highly sought-after manager attribute imply that in the absence of a manager rating, individuals believe that male managers are better mentors than female managers.

6.2 Heterogeneity

6.2.1 Distribution of Preference and Belief Parameters We leverage our panel structure of individual data to estimate the model for each individual separately to obtain individual-specific estimates of preference and belief parameters. Raw individual-specific estimates reveal substantial heterogeneity: approximately 64% of workers prefer working for female managers, while 61% believe female managers are worse mentors when mentorship information is unavailable. However, these individual-level estimates are noisy—each set being estimated from 20 scenarios. To avoid overstating the variance in the distribution obtained from noisy estimates, while accounting for heterogeneous standard errors across workers, we implement a non-parametric empirical Bayes (NPEB, henceforth) shrinkage procedure.³¹ We estimate the underlying distribution of the parameters using log-spline deconvolutions following [Efron \(2016\)](#) and discussed in [Walters \(2024\)](#), and account for the varying precision of individual estimates through their respective standard errors to produce precision-adjusted posterior means (see [Alam et al. \(2026b\)](#) Online Appendix C for details).

The distributions of belief and preference parameters obtained from the NPEB estimates are plotted in Figure 3, Panels A and B respectively, overlaid on top of the distributions obtained from the raw estimates. While variance is reduced as expected, the key patterns discussed above persist. Absent information on mentorship, 66.55% of workers believe that female managers are worse mentors absent information, and 75.86% of workers prefer female managers in the presence of information on mentorship quality.

³⁰In addition, note that $\frac{\mathbb{E}(\tilde{\beta}_i^G - \beta_i^G)}{\mathbb{E}(\beta_i^R)} \neq \mathbb{E}\left(\frac{\tilde{\beta}_i^G - \beta_i^G}{\beta_i^R}\right) = \mathbb{E}(\alpha_i^G)$. However, $\mathbb{E}(\tilde{\beta}_i^G - \beta_i^G) = \mathbb{E}(\beta_i^R \alpha_i^G)$ provides the average valuations of beliefs.

³¹Unlike parametric shrinkage methods, the non-parametric approach allows the underlying distribution to exhibit arbitrary features such as multimodality or skewness.

6.2.2 Correlates of Parameters with Observable Characteristics To examine whether the heterogeneity in discrimination parameters correlates systematically with worker characteristics, we employ a simple machine learning approach. We use LASSO (Least Absolute Shrinkage and Selection Operator) to select the most relevant predictors from our full set of demographic characteristics including gender, field of study, family income, parental education levels, parental employment status and their interactions. This approach has the advantage of reducing the dimensionality of demographics and thereby circumventing concerns about multiple hypothesis testing which would arise while directly testing associations with numerous demographic variables.³² Following variable selection, we regress the NPEB parameter estimates on the LASSO-selected demographics to document how they correlate with observable demographics.

Table 8 presents the results. For beliefs about the gender gap in mentorship, LASSO selects three variables: a parental education combination (those with fathers holding master's and mothers above-master's degree), own education (those with advanced degrees i.e., MPhil and PhD) and father's employment status. OLS estimates of NPEB belief estimates on LASSO-selected variables reveal that respondents pursuing education beyond a master's degree are less likely to hold negative beliefs about female managers' mentoring ability. We also find individuals whose mothers are more educated than their fathers—specifically those with fathers holding master's and mothers above-master's degree—are less likely to hold negative beliefs about female managers' mentoring ability. Those whose fathers are unemployed also are less likely to hold such beliefs, though this association is approximately one-third the magnitude of both the education associations.³³ For preferences over manager gender, LASSO selects no demographic variables. The absence of systematic demographic predictors for preferences, despite substantial individual-level heterogeneity, suggests that preferences for manager gender are driven by idiosyncratic factors beyond observable characteristics.

Exposure to higher education is associated with a reduced likelihood of holding such beliefs, consistent with education reducing biased beliefs in other contexts, for example immigrants (Dylong and Uebelmesser, 2024) and inflation (D'Acunto et al., 2023). When mothers attain higher education than fathers within a household, especially with advanced degrees, this plausibly signals high gender-specific abilities that counteract biased stereotypes about gender.

7. OTHER MANAGERIAL QUALITIES AND POTENTIAL MECHANISMS

Since our main results show that mentorship is highly valued, understanding the source of these beliefs is crucial. To study this, we consider additional gender-specific managerial traits, and extend the model of beliefs about mentorship by allowing it to also

³²The LASSO estimator solves $\min_{\beta} \frac{1}{2n} \sum_{i=1}^n (\hat{\theta}_i^{NPEB} - x_i' \beta)^2 + \lambda \sum_{k=1}^K |\beta_k|$, where $\hat{\theta}_i^{NPEB}$ represents the shrunk estimates, x_i represents K -dimensional vector of individual characteristics, and λ denotes the penalty parameter. We determine the optimal λ via 10-fold cross-validation over a grid of 100 values, that minimizes mean squared prediction error.

³³The mechanism underlying this association remains unclear without additional information about fathers' workplace experiences, that may be transmitted to their children.

depend on other traits. This allows us to (a) quantify the weight respondents place on those traits in inferring mentorship quality in the incomplete scenarios, (b) quantify the share of the gender gaps in beliefs about mentorship arising through perceptions about gender gaps in other traits, (c) examine whether the beliefs about the gender gap in mentorship persist after conditioning on these traits and (d) thereby explore mechanisms driving our main results. To this end, we collect additional data on individuals' perceptions of three other managerial qualities: competence, pleasantness, and non-discriminatory behavior where individuals report ordinal data on whether they perceive males to be better, similar, or worse than females in each of these traits.

In Appendix E we extend our model to incorporate these additional traits, and formally show a three-step semi-parametric identification using cross-sectional variation across reported perceived traits in addition to within individual variation across complete and incomplete scenarios in Appendix E.3. Estimation and inference similarly follows a three-step joint block-bootstrap process explained in Appendix E.4.

We document four key findings from the estimates of the extended model reported in Appendix Table G and explained in more detail in Appendix E.5. First, individuals perceive that male managers are substantially more competent than female managers.³⁴ Second, we find that competence significantly predicts mentorship ability, while pleasantness and non-discriminatory behavior show no significant relationship with mentorship quality (Appendix Table G: Panel B, column 2). Third, we estimate the beliefs about the gender gap in mentorship conditional on other managerial traits and compare them to our baseline estimates to understand how much the mentorship belief gap reduces conditional on other managerial traits. Specifically, suppressing other job attributes to reduce notational clutter, we compare estimates of $\mathbb{E}[\beta_i^R R_j \mid G_j = F, \mathbb{Q}] - \mathbb{E}[\beta_i^R R_j \mid G_j = M, \mathbb{Q}]$ from our extended model where $\mathbb{Q}$ denotes the vector of other qualities, to our baseline estimates of $\mathbb{E}[\beta_i^R R_j \mid G_j = F] - \mathbb{E}[\beta_i^R R_j \mid G_j = M]$. While qualitatively similar, the beliefs about the mentorship gap once conditioned on other managerial traits reduce to 0.6% of average annual wages (Appendix Table G: Panel C, column 1). This is a 62.5% reduction in comparison to our estimates from the baseline model. Fourth, we decompose the total valuation of the beliefs about the mentorship gap and find that 43.6% of the mentorship gap is driven by perceptions of gender-based trait differences (Appendix Table G: Panel C, column 2).

An assumption underlying this decomposition is that respondents' reported perceptions of other managerial traits reflect their true perceptions. If social desirability concerns lead individuals to overstate the quality of females on traits, our estimates of gender gaps in these traits could be attenuated. In such a case, our estimate that 43.6% of the mentorship belief gap is explained by perceived traits is a lower bound. With workers valuing one SD of mentorship at 5.4% of wages, if their beliefs representing a 0.29 SD gender gap in mentorship—at least half of which are driven by perception that women are less competent—are inaccurate, they could create substantial welfare losses.

³⁴See Appendix Table G: Panel B, column 1 and Appendix Figure G.2 for the estimated distributions in the perceived gender gap of the three latent traits.

8. ARE NEGATIVE BELIEFS ABOUT FEMALE MENTORSHIP BIASED?

Our findings reveal that workers believe female managers are worse mentors than male managers, absent information about mentorship quality. While we lack data on the population distribution of mentorship quality by gender in India, extensive evidence from other contexts consistently documents that female mentors are at least as effective as male mentors, and often more so, especially for female mentees. Early meta-analyses find economically negligible gender differences in mentoring effectiveness (O'Brien et al., 2010) and observed gaps reflect differences in resources available to mentors rather than mentoring ability itself (Schwartz et al., 2022). Experimental and quasi-experimental evidence from diverse settings—entrepreneurship (Germann et al., 2024), corporate settings (Cardoso and Winter-Ebmer, 2007, Kunze and Miller, 2017, He and le Maire, 2022), academia (Carrell et al., 2010, Canaan and Mouganie, 2021, 2023, Kjelsrud and Parsa, 2024, Blau et al., 2010, Ginther and Na, 2021)—consistently corroborates this conclusion. Consistent with this body of evidence, Weidmann et al. (2024) document that gender is not a significant predictor of overall managerial effectiveness and Alfonsi et al. (2022) finds no difference in career trajectories between mentees of male and female mentors. These findings suggest that the negative beliefs we document are likely biased. Moreover, biased beliefs about group characteristics in general are a well-documented and pervasive phenomenon across diverse economic contexts.³⁵ Assuming workers hold correct beliefs when they do not can lead to suboptimal policy design (Conlon et al., 2018). This matters quantitatively in our context because of workers' strong preference for high mentorship quality. Hence, if beliefs are biased, that would potentially generate substantial welfare loss through suboptimal matching.

9. POLICY IMPLICATIONS

Our findings, combined with evidence that female mentors are at least as effective as male mentors, suggest a simple policy intervention of disclosure of manager ratings. Many firms already collect such data internally. Making these ratings transparent could correct any biased beliefs, and suggestively improve job matching efficiency. Our heterogeneity results indicate targeting of such policies towards workers without advanced degrees and those from households with traditional educational differences (fathers more educated than mothers) since they are more likely to hold negative beliefs about female managers' mentorship ability. Additionally, as female mentors are more effective when paired with female mentees, such information provision interventions could be suggestively more beneficial for female workers.

Such information provision could also suggestively enhance efficacy of mentoring programs in general. Potentially biased beliefs imply mentees under-match with female

³⁵Documented evidence includes misperceptions about returns to education (Wiswall and Zafar, 2015), immigration levels and immigrant characteristics (Alesina et al., 2019, Dylong and Uebelmesser, 2024), social norms about female labor force participation (Bursztyn et al., 2020), economic indicators (D'Acunto et al., 2023), income distributions (Hvidberg et al., 2020), college returns (Bleemer and Zafar, 2018), gender gaps in social preferences (Exley et al., 2025), discrimination (Bohren et al., 2019), and racial and diversity statistics (Alesina and Stantcheva, 2020).

mentors, limiting program effectiveness. Providing aggregated mentee outcome statistics such as career progression would serve as credible signals of mentorship quality to correct potentially biased beliefs. Additionally, given our finding that workers perceive that competence and mentorship ability are positively correlated, visible mentee success statistics would suggestively also signal mentor competence more broadly.

10. VALIDATION EXERCISES

10.1 *Validation of indirect information experiment approach*

After our information experiment, we directly elicited beliefs about manager mentorship which shows that direct elicitation methods are susceptible to social desirability bias while studying sensitive topics like gender, race, corruption, discrimination etc., consistent with existing evidence (Bursztyn et al., 2025, Björkman Nyqvist et al., 2018, Hainmueller and Hangartner, 2013, Kuklinski et al., 1997, Kraay and Murrell, 2016).

Respondents evaluated 10 jobs with exogenously varied manager names, wages, and flexible hours (see Appendix Figure G.3). For each job, they reported their expected manager mentorship rating on a zero-to-five scale. Using these data we estimate $R_{ij}^{direct} = \theta_i + \alpha^G G_j + \alpha^H H_j + \alpha^W W_j + \eta_{ij}$ where R_{ij}^{direct} is individual i 's directly elicited mentorship rating for the manager in job j , G_j indicates a male manager, H_j indicates availability of flexible hours, W_j are annual wages, and θ_i is an individual fixed effect. Appendix Table G.4 presents the results. Directly elicited beliefs show the opposite pattern from our model-based estimates. Respondents when directly asked, on average believe female managers are better mentors ($\hat{\alpha}^G = -0.042$, statistically significant at 1%). This is equivalent to -0.08SD in beliefs about the gender gap in mentorship. This reversal is driven by male respondents ($\hat{\alpha}^G = -0.053$, statistically significant at 5%), while female respondents' directly elicited beliefs about the gender gap in mentorship are statistically indistinguishable from zero.

This reversal—from negative beliefs revealed in the information experiment to positive beliefs about female mentorship when asked directly—validates our information experiment approach for studying beliefs. Furthermore, our model-based estimates are consistent with the raw choice data patterns where in the presence of information on mentorship, jobs with female managers are chosen at substantially higher rates than in the absence of information (by 18.0% from 51.8% in incomplete scenarios to 61.1% in complete scenarios), suggesting negative beliefs about mentorship. While learning during our experiment could potentially influence directly elicited beliefs, our design rules this out as the primary explanation. Since we balanced mentorship ratings across manager genders, learning would lead to no gender gap in directly elicited beliefs. Instead, we observe a significant reversal favoring female managers, particularly among male respondents, pointing to social desirability bias as the primary explanation. This underscores the value of indirect elicitation methods for socially sensitive topics like gender, race, discrimination, and immigration etc.

In other contexts where behaviors are less normatively sensitive or where social-image concerns are minimal, direct elicitation methods have been found to be reliable.

For example, survey-based reports of educational attainment closely align with administrative records (Lamb Jr and Stem Jr, 1978, Kleven and Ringdal, 2020). Similarly, in the context of voting in Norway, a 96 percent match exists between self-reported turnout and registry data (Kleven, 2022), suggesting that in high-trust settings with strong civic norms and minimal perceived judgment, direct elicitation can be accurate. In such cases, direct elicitation methods should be preferred to avoid the cost of conducting information experiments.

10.2 *Validation of using compensating differentials*

The cognitive load of reporting compensating differentials could be plausibly higher than reporting a choice. Since preferences are identified from the complete scenarios using either compensating differentials or standard discrete choice methods, we can compare the estimates from these two approaches to provide a sanity check of our survey design and usage of compensating differentials.

Table 9 presents estimates, standard errors and 95% confidence intervals of willingness to pay (WTP) estimates from both approaches as percent of average annual wages. Both approaches yield qualitatively similar estimates that are economically close in magnitude, providing a validation of our survey design. The conditional logit yields a WTP for male managers of -3.3% of average annual wages, while our compensating differential approach yields -1.7% with 95% confidence intervals of $[-4.16, -2.45]$ and $[-2.21, -1.15]$ respectively. While the confidence intervals do not overlap, the economic magnitude of the difference is modest. The two approaches yield statistically indistinguishable estimates for flexible hours. Despite differing for mentorship rating, importantly for the identification of beliefs, both approaches indicate strong preference for managers with higher mentoring quality. Finally, note that our compensating differentials approach yields more conservative estimates of the willingness to pay parameters relative to those obtained from standard discrete choice methods. While both methods identify preferences, identification of beliefs necessitates the use of our compensating differentials approach.

11. CONCLUSION

We develop a general identification framework for recovering the distributions of preferences and beliefs in settings where agents choose among options with unobserved attributes to maximize expected utility whereby their choices conflate preferences with beliefs about those attributes. We formally show that compensating differentials separately identify these distributions, whereas choice probabilities do not in such settings. The key insight is that compensating differentials, being monetary valuations, remain invariant across information structures, whereas choice probabilities fail due to non-innocuous variance normalizations. Identification of preferences exploits variation in compensating differentials within complete information scenarios, while beliefs are identified from variation in compensating differentials between complete and incomplete information scenarios.

We apply our method to provide the first evidence on the distributions of worker preferences for manager gender and beliefs about mentorship ability. By systematically varying the availability of manager mentorship information in hypothetical job choices between complete and incomplete scenarios in within-individual information experiments, we separately identify beliefs from preferences. We find that workers prefer female managers when fully informed, willing to forgo 1.7% of average annual wages. However, in the absence of information 67% of workers believe female managers are worse mentors. The average belief-driven discrimination is equivalent to 1.6% of wages and offsets their preferences in the absence of information. In a separate exercise, we find strong evidence that direct elicitation of beliefs suffers from social desirability bias, particularly among male respondents. This shows that our indirect approach to identify beliefs are particularly useful in settings where social desirability bias is of concern.

These beliefs are consequential given that workers value mentorship highly, willing to forgo 5.4% of wages for a one-standard-deviation improvement in mentorship quality. Yet 67% believe female managers provide worse mentoring, a belief worth 1.6% of wages that offsets their underlying preference for female managers. These beliefs are primarily driven by the perception that female managers are less competent, and that competence is a key predictor of mentorship effectiveness, and are prevalent among workers without advanced degrees and those from households with traditional educational hierarchies. A large body of evidence suggests that these beliefs are biased indicating the importance of information-based policy interventions.

As labor markets increasingly recognize the value of mentorship—with firms investing millions in mentoring initiatives while struggling to break glass ceilings—understanding how beliefs about gender gaps in mentorship shape workers' sorting decisions is essential for designing effective policies to reduce disparities and improve match efficiency.

More broadly, identification of subjective beliefs is particularly important given that subjective beliefs do not always align with typically assumed rational expectations (see e.g., [Kapor et al. \(2020\)](#), [Jäger et al. \(2021b\)](#)). Beyond this application, our framework provides an identification strategy for any setting where agents choose among options with unobserved attributes to maximize expected utility and information experiments are feasible.

12. TABLES AND FIGURES

FIGURE 1. Schematic representation of survey flow

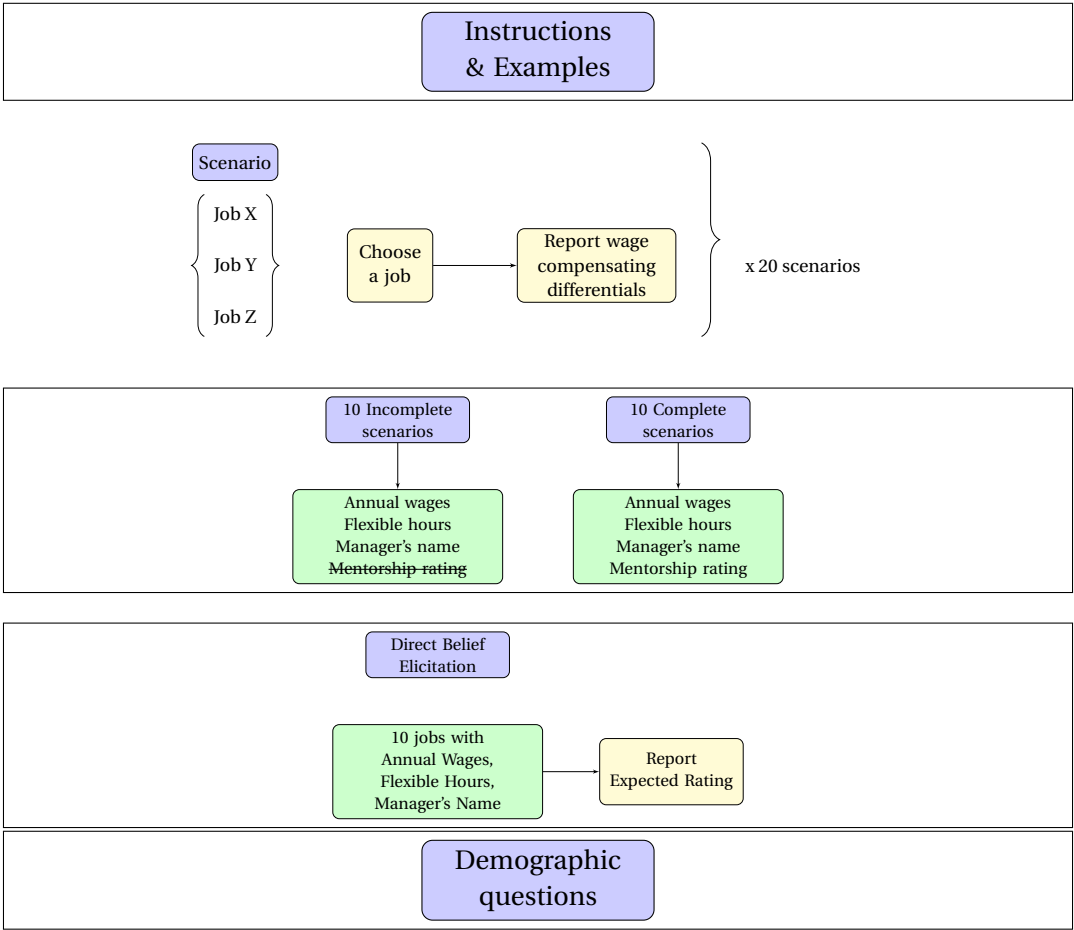


FIGURE 1. *Notes:* Every scenario consists of three different jobs: X, Y and Z. Individuals choose their most preferred job. For the jobs that they do not choose, individuals are asked to report the minimum increase in wages that they would need to choose those jobs instead. There are 20 such scenarios. In the first 10 scenarios, individuals do not observe the manager mentorship rating; however, in the last 10, they do, along with the other attributes.

TABLE 1. Incomplete scenario example

Job choice

Rating of each manager could be different, but the data is unavailable. Anything else that you don't see here, is the SAME across all jobs. Please select your most preferred job.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	6	6.6	6.4
Flexible hours	yes	no	yes
Manager	Anirban	Shrinita	Arup
Manager rating	N/A	N/A	N/A

Job X

Job Y

Job Z

Compensating differential (if job chosen was Y)

You chose Job Y.

If you were to negotiate your wage, how much of a MINIMUM INCREASE IN WAGES would you need in each of the other jobs for you to choose it instead of Job Y? The scale here ranges from 0 to 2 lakhs.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	6	6.6	6.4
Flexible hours	yes	no	yes
Manager	Anirban	Shrinita	Arup
Manager rating	N/A	N/A	N/A

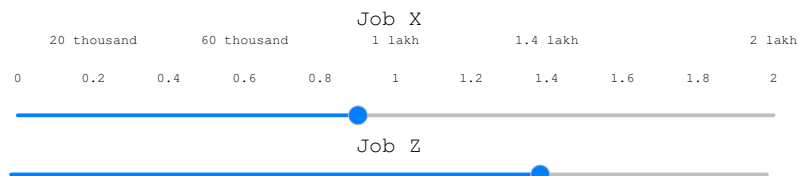


TABLE 1. *Notes:* Jobs X and Z have male managers and Job Y has a female manager. Across the 10 incomplete scenarios, five scenarios have two jobs with male managers and one job with a female manager, and the remaining five have two jobs with female managers and one job with a male manager.

TABLE 2. Complete scenario example

Job choice

Anything else that you don't see here, is the SAME across all jobs. Manager rating is on a scale of 1-5. 1: Poor; 2: Fair; 3: Good; 4: Very good; 5: Excellent. Please select your most preferred job.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	8.2	8	8.6
Flexible hours	yes	no	yes
Manager	Mohan	Mohit	Mahima
Manager rating	3.70	4.00	3.15

Job X

Job Y

Job Z

Compensating differential example (if job chosen was X)

You chose Job X.
If you were to negotiate your wage, how much of a MINIMUM INCREASE IN WAGES would you need in each of the other jobs for you to choose it instead of Job X?
The scale here ranges from 0 to 2 lakhs.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	8.2	8	8.6
Flexible hours	yes	no	yes
Manager	Mohan	Mohit	Mahima
Manager rating	3.70	4.00	3.15

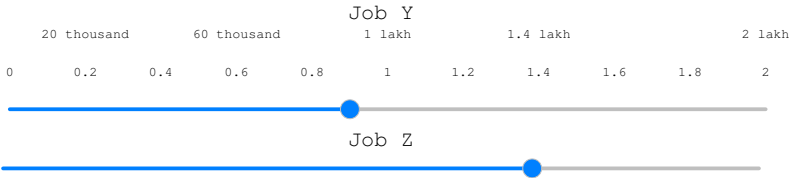


TABLE 2. Notes: Jobs X and Y have male managers, and Job Z has a female manager. Overall, across the 10 complete scenarios, five scenarios have two jobs with male managers and one job with a female manager, and the remaining five have two jobs with female managers and one job with a male manager.

TABLE 3. Summary statistics of job attributes across managers and scenarios

Panel A: Summary statistics of job attributes

Variable	Mean	Std. Dev.	N
Female Manager	0.50	0.504	60
Flexible hours	0.53	0.503	60
Annual wages	7.11	1.476	60
Rating	3.41	0.495	30

Panel B: Job attributes across male and female managers

Attribute	Male Manager	Female Manager	Difference	p-value
Rating (only complete scenarios)	3.373 (0.519) [0.134]	3.447 (0.485) [0.125]	0.073 [0.183]	0.692
Flexible hours	0.567 (0.504) [0.092]	0.500 (0.509) [0.093]	-0.067 [0.131]	0.612
Annual wages	7.080 (1.529) [0.279]	7.140 (1.445) [0.264]	0.060 [0.384]	0.876

Panel C: Job attributes across male and female managers by scenario type*Incomplete scenarios*

Attribute	Male Manager	Female Manager	Difference	p-value
Flexible hours	0.600 (0.507) [0.131]	0.533 (0.516) [0.133]	-0.067 [0.187]	0.724
Annual wages	7.373 (1.533) [0.396]	7.127 (1.518) [0.392]	-0.247 [0.557]	0.661

Complete scenarios

Attribute	Male Manager	Female Manager	Difference	p-value
Flexible hours	0.533 (0.516) [0.133]	0.467 (0.516) [0.133]	-0.067 [0.189]	0.726
Annual wages	6.787 (1.520) [0.392]	7.153 (1.423) [0.367]	0.367 [0.538]	0.501
Rating	3.373 (0.519) [0.134]	3.447 (0.485) [0.125]	0.073 [0.183]	0.692

TABLE 3. *Notes:* Panel A shows summary statistics of job attributes in the 60 jobs shown to respondents (3 jobs per scenario across 10 incomplete and 10 complete scenarios). The mentorship rating statistics come from the last 30 jobs because they are only shown in the 10 complete scenarios. Panel B shows the balance of job attributes by manager gender across all scenarios. Panel C shows the balance separately for incomplete and complete scenarios. Numbers in parentheses contain standard deviations and numbers in square brackets contain standard errors.

TABLE 4. Sample demographics

	Gender		Total
	Male	Female	
580 respondents	58.3	41.7	100
Area of Study:			
Arts	25.1	71.9	44.7
Engineering	49.4	10.7	33.3
Science	25.4	17.4	22.1
Family Income:			
Less than 2 lakhs (Less than \$9,492)	37.6	24.4	32.1
2 lakhs to 5 lakhs (\$9,492 to \$23,730)	26.9	24.0	25.7
5 lakhs to 10 lakhs (\$23,730 to \$47,460)	21.3	30.2	25.0
10 to 20 lakhs (\$47,460 to \$94,921)	11.5	15.3	13.1
Above 20 lakhs (Above \$94,921)	2.7	6.2	4.1
Mother's Education:			
Below High School	17.8	11.2	15.0
High School	32.8	18.6	26.9
Bachelor's	38.5	46.3	41.7
Master's	8.0	16.9	11.7
Above Master's	3.0	7.0	4.7
Father's Education:			
Below High School	9.2	7.4	8.4
High School	21.3	11.2	17.1
Bachelor's	51.5	55.4	53.1
Master's	13.9	17.8	15.5
Above Master's	4.1	8.3	5.9
Mother's Occupation:			
Government	10.9	15.3	12.8
Homemaker	70.1	63.2	67.2
Not Applicable	4.4	3.3	4.0
Private Sector	4.7	8.7	6.4
Self-Employed	9.8	9.5	9.7
Father's Occupation:			
Government	30.8	33.1	31.7
Homemaker	3.3	0.8	2.2
Not Applicable	16.3	11.2	14.1
Private Sector	16.9	20.7	18.4
Self-Employed	32.8	34.3	33.4

TABLE 4. *Notes:* All variables are categorical. Numbers represent percentages. The parental occupation category of “Not Applicable” refers to a deceased parent. Variables on income categories are in INR and have their corresponding purchasing power parity–adjusted USD equivalent below each category.

TABLE 5. Job choices and compensating differentials by manager gender and scenario type

Panel A: Percentages of chosen jobs with male and female managers

<i>Respondent</i>		<i>All</i>		<i>Female</i>		<i>Male</i>	
<i>Scenario</i>	Manager	Male	Female	Male	Female	Male	Female
Incomplete		48.2	51.8	48.1	51.9	48.3	51.7
Complete		38.9	61.1	38.8	61.2	38.9	61.1

Panel B: Average compensating differentials for unchosen jobs

	Male Manager	Female Manager	Difference	St. Diff.
Incomplete scenarios	0.957 (0.965)	1.020 (0.998)	-0.063 (0.018)	-0.046
Complete scenarios	1.100 (1.058)	1.038 (0.977)	0.061 (0.019)	0.042

TABLE 5. *Notes:* Panel A shows the percent of jobs chosen with male and female managers in the incomplete scenarios (where mentorship rating was not shown to respondents) and in the complete scenarios (where mentorship rating was shown). Percentages are shown for all respondents and disaggregated by respondent gender. Panel B shows average compensating differentials demanded by respondents for unchosen jobs, in units of 1 lakh (hundred thousand) INR. Standard deviations in parentheses, standard errors for differences in parentheses.

TABLE 6. Difference-in-differences estimates from the complete scenarios data

VARIABLES	Margins
Female Worker	-0.001 (0.007)
Female Manager	0.081 (0.010)
Female Worker X Female Manager	0.001 (0.014)
Annual Wages	0.347 (0.022)
Mentorship Rating	0.484 (0.016)
Flexible Hours	0.262 (0.013)
Scenario FE	yes
Observations	17,400

TABLE 6. *Notes:* The estimates show the marginal effects of each of the attributes in the difference-in-differences specification (1). Standard errors bootstrapped at the individual level with 1,000 replications. The total number of observations is 17,400 because we use individual-level choice data on 3 jobs in each of the 10 complete scenarios for 580 individuals.

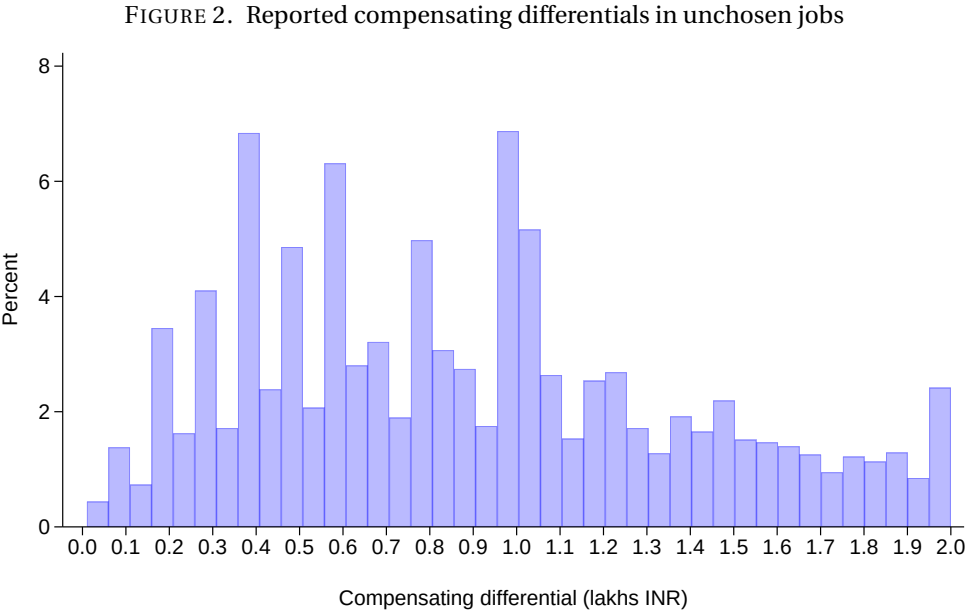


FIGURE 2. *Notes:* The increase in wages is in units of 1 lakh (hundred thousand) INR. The figure is plotted for values only between 0 and 2 lakhs.

TABLE 7. Preference and Belief Parameter Estimates

Panel A: Complete and Incomplete Scenarios: Jointly Estimated					
Incomplete Scenarios			Complete Scenarios		
Parameters	in 10 ⁵ INR	% of wages	Parameters	in 10 ⁵ INR	% of wages
$\tilde{\beta}_i^G = \beta_i^G + \beta_i^R \alpha_i^G$	-0.007 (0.012)	0.1%	β_i^G (Male Manager)	-0.119 (0.019)	1.7%
$\tilde{\beta}_i^H = \beta_i^H + \beta_i^R \alpha_i^H$	1.134 (0.059)	16.0%	β_i^H (Flexible Hours)	0.777 (0.027)	10.9%
$\tilde{\beta}_i^W = \beta_i^W + \beta_i^R \alpha_i^W$	1.369 (0.068)	19.3%	β_i^W (Annual Wages)	1	
			β_i^R (Mentorship)	0.792 (0.030)	11.1%
Observations	11,600			11,600	
Panel B: Belief Parameters					
Belief Parameter		Valuation of Beliefs			
	Estimate			in 10 ⁵ INR	% of wages
α_i^G (Male managers)	0.142 (0.029)		$\beta_i^R \alpha_i^G$ (Male managers)	0.112 (0.023)	1.6%

Notes: Panel A shows estimates from estimating equation system (16) for each individual and reports the averages $E(\tilde{\beta}_i^x)$ for each attribute $x \in \{G, H, W\}$ in the incomplete scenarios and $E(\beta_i^x)$ for each attribute $x \in \{G, H, R\}$ in the complete scenarios. β_i^W is normalized to 1. Panel B shows belief parameters where $\alpha_i^G = \frac{\tilde{\beta}_i^G - \beta_i^G}{\beta_i^R}$ and the valuation of beliefs $\beta_i^R \alpha_i^G = \tilde{\beta}_i^G - \beta_i^G$. Estimates are represented in two sets of units—the first is in hundred thousand INR, and the second converts those units into percentages of average annual wages. Standard errors are computed using the block bootstrap at the individual level with 1,000 replications. Average annual wages equal 7.11 lakh INR ($\approx$ \$33.7 thousand in PPP).

TABLE 8. Heterogeneity in Parameters by LASSO-selected demographics

	(1) Valuation of Beliefs	(2) Preferences
<i>Father's × Mother's education</i>		
Master's × Above Master's	-0.117 (0.063)	
<i>Own education</i>		
Mphil/PhD	-0.100 (0.045)	
1 (Father employed)=0	-0.038 (0.016)	
Constant	0.074 (0.006)	-0.076 (0.005)
Observations	580	580
R-squared	0.024	0.000

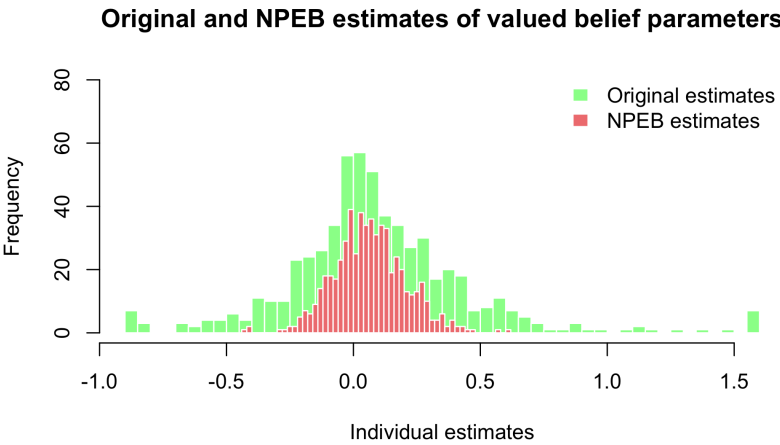
Notes: Outcome variables are estimates obtained after shrinking individual-level structural parameter estimates on preferences and beliefs using a non-parametric empirical Bayes procedure. The preference parameter captures willingness to forgo wages to work for male managers. The belief parameter captures beliefs about male managers' mentorship ability relative to female managers. Demographic variables are selected using LASSO (Least Absolute Shrinkage and Selection Operator) with 10-fold cross-validation from a set of demographic covariates including gender, field of study, education level, family income, parental education, parental employment, and interactions of parental demographics. The penalty parameter λ was chosen from a grid of 100 values to minimize mean squared error.

TABLE 9. Validation: Comparison of WTP Estimates from Alternative Approaches

Attribute	Discrete Choice	Compensating Differential
Male Manager	-3.3% (0.44) [-4.16, -2.45]	-1.68% (0.27) [-2.21, -1.15]
Flexible Hours	10.62% (0.77) [9.11, 12.13]	10.92% (0.37) [10.19, 11.66]
Mentorship Rating (SD units)	9.54% (0.58) [8.4, 10.67]	5.42% (0.2) [5.02, 5.82]

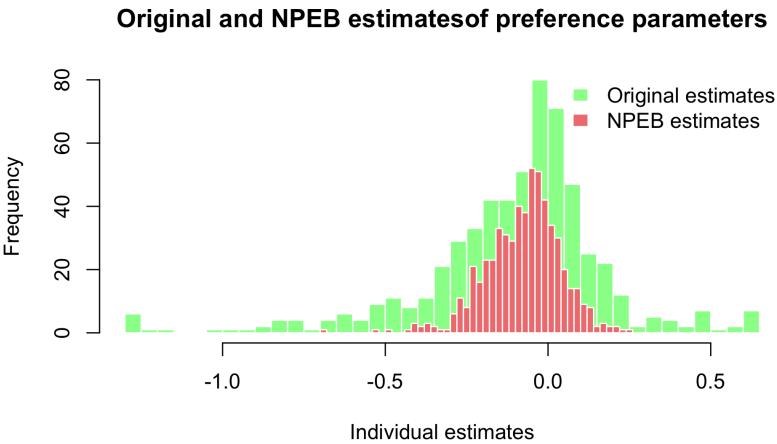
TABLE 9. *Notes:* The discrete choice estimates are from a conditional logit model on choices in the 10 complete scenarios with individual and scenario fixed effects and standard errors clustered at the individual level. The compensating differential estimates are from the structural model using data from complete scenarios. WTP is expressed as a percentage of average annual wages (≈ 7.11 lakhs INR). Standard errors in parentheses and 95% confidence intervals in square brackets.

FIGURE 3. Distribution of Individual Raw Estimates and NPEB Estimates
(a) Beliefs on Male relative to Female Managers' Mentorship



Notes: The figure plots the histograms of the original individual estimates of the valuation of beliefs about male managers' mentorship ability relative to female managers (winsorized at the 1st and 99th percentiles), and overlays the corresponding non-parametric empirical Bayes (NPEB) estimates.

(b) Preferences on Male relative to Female Managers



Notes: The figure plots the histograms of the original individual estimates of the preference for male managers relative to female managers (winsorized at the 1st and 99th percentiles), and overlays the corresponding non-parametric empirical Bayes (NPEB) estimates.

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APPENDIX A: APPENDIX: IDENTIFICATION

Let individual $i \in \{1, \dots, N\}$ have preferences over attributes X_j in job $j \in \{1, \dots, J\}$ given by the function

$$U_{ij} = u_i(X_j) + \epsilon_{ij}$$

Identification is achieved in two steps. We first show the identification of preferences when information is complete in Step 1. Then, with incomplete information in Step 2, we show that we can identify beliefs given preferences.

Step 1: Identifying preferences

Individuals form expectations when they report their job choices and compensating differentials. Hence,

$$\mathbb{E}_i[U_{ij} | X_j] = u_i(X_j) + \mathbb{E}_i[\epsilon_{ij} | X_j]$$

A compensating differential of Δ_{ijk} makes i indifferent between jobs j and k when i observes X_j and X_k . Hence, given that a compensating differential increases only wages and changes nothing else about the job, we have $\mathbb{E}_i[\epsilon_{ik} | X_k, \Delta_{ijk}] = \mathbb{E}_i[\epsilon_{ik} | X_k]$. Normalizing the preference parameter on wages to 1, we have

$$\begin{aligned} \mathbb{E}_i[U_{ik} | X_k, \Delta_{ijk}] &= \Delta_{ijk} + u_i(X_k) + \mathbb{E}_i[\epsilon_{ik} | X_k, \Delta_{ijk}] \\ &= \Delta_{ijk} + u_i(X_k) + \mathbb{E}_i[\epsilon_{ik} | X_k] \end{aligned}$$

Given that everything else about the job is the same, we have $\mathbb{E}_i[\epsilon_{ik} | X_k] = \mathbb{E}_i[\epsilon_{ij} | X_j]$ for all jobs $j \neq k$.

Since by definition $\Delta_{ijk} = \mathbb{E}_i[U_{ij} | X_j] - \mathbb{E}_i[U_{ik} | X_k, \Delta_{ijk}]$, we now have

$$\Delta_{ijk} = u_i(X_j) - u_i(X_k)$$

As the number of scenarios goes to infinity, for each individual i , this identifies preferences $u_i(\cdot)$ as long as $\text{Var}_i(u_i(X_j) | X_j) \neq 0$.

Step 2: Identifying beliefs given preferences

Now, when i observes $\tilde{X}_j \equiv X_j \setminus R_j$ for each job $j \in \{1, \dots, J\}$, i forms beliefs about R_j given $\tilde{X}_j$ according to $G_i(R | \tilde{X})$. Now, we have

$$\mathbb{E}_i[U_{ij} | \tilde{X}_j] = \mathbb{E}_i[u_i(X_j) | \tilde{X}_j] + \mathbb{E}_i[\epsilon_{ij} | \tilde{X}_j]$$

The expectation here varies by individuals. This allows different individuals to draw from different distributions of mentorship, which may not necessarily be the true distribution. A compensating differential of $\tilde{\Delta}_{ijk}$ makes i indifferent between jobs j and k while observing $\tilde{X}_j$ and $\tilde{X}_k$, respectively. Hence, given that the compensating differential increases only wages and changes nothing else about the job, and normalizing the

preference parameter on wages to 1, we have

$$\begin{aligned}\mathbb{E}_i[U_{ik} \mid \widetilde{X}_k, \widetilde{\Delta}_{ijk}] &= \widetilde{\Delta}_{ijk} + \mathbb{E}_i[u_i(X_k) \mid \widetilde{X}_k] + \mathbb{E}_i[\epsilon_{ik} \mid \widetilde{X}_k, \widetilde{\Delta}_{ijk}] \\ &= \widetilde{\Delta}_{ijk} + \mathbb{E}_i[u_i(X_k) \mid \widetilde{X}_k] + \mathbb{E}_i[\epsilon_{ik} \mid \widetilde{X}_k]\end{aligned}$$

Given that everything else about the job is assumed to be the same, we have $\mathbb{E}_i[\epsilon_{ik} \mid \widetilde{X}_k] = \mathbb{E}_i[\epsilon_{ij} \mid \widetilde{X}_j]$ for all jobs $j \neq k$.

Since by definition $\widetilde{\Delta}_{ijk} = \mathbb{E}_i[U_{ij} \mid \widetilde{X}_j] - \mathbb{E}_i[U_{ik} \mid \widetilde{X}_k, \widetilde{\Delta}_{ijk}]$, we have

$$\begin{aligned}\widetilde{\Delta}_{ijk} &= \mathbb{E}_i[u_i(X_j) \mid \widetilde{X}_j] - \mathbb{E}_i[u_i(X_k) \mid \widetilde{X}_k] \\ &= \int u_i(X_j) dG_i(R \mid \widetilde{X} = \widetilde{X}_j) - \int u_i(X_k) dG_i(R \mid \widetilde{X} = \widetilde{X}_k)\end{aligned}$$

Compact support of R and $\widetilde{X}$ ensures that moments exist. As the number of scenarios goes to infinity, the above non-parametrically identifies $\mathbb{E}_i[u_i(\widetilde{X}_j, R_j) \mid \widetilde{X}_j]$ using variation in $\widetilde{\Delta}_{ijk}$ based on exogenously varied $\widetilde{X}_j$ and $\widetilde{X}_k$. This requires $\text{Var}_i[u_i(X_j) \mid \widetilde{X}_j] \neq 0$, which ensures that unobserved R affects utility. Combined with $u_i(\cdot)$ non-parametrically identified from Step 1, this recovers moments of the belief distribution $G_i(R \mid \widetilde{X})$ that enter expected utility.¹ When $u_i(\cdot)$ is sufficiently flexible in R to generate a complete family of moment conditions, this identifies $G_i(R \mid \widetilde{X})$.

Thus, variation between $\widetilde{\Delta}_{ijk}$ and Δ_{ijk} based on exogenously varied $\widetilde{X}_j$ and $\widetilde{X}_k$ across complete and incomplete scenarios non-parametrically identifies the belief function that enters expected utility in the absence of information on R . The model is non-parametrically identified as the number of parameters characterizing preferences and beliefs goes to infinity as long as it approaches infinity at a slower rate than the number of scenarios approaches infinity.

Remark: Non-identification

The belief parameters are not identified under two circumstances.

1. If $u_i(X_j)$ does not depend on R , then choices and compensating differentials are uninformative about beliefs. To see this mathematically, the above set of equations become independent of $G_i(R \mid \widetilde{X})$ because $\mathbb{E}_i[u_i(X_j) \mid \widetilde{X}_j] = \mathbb{E}_i[u_i(\widetilde{X}_j) \mid \widetilde{X}_j] = u_i(\widetilde{X}_j)$. The first equality follows from $\widetilde{X}_j \equiv X_j \setminus R_j$, and that i does not care about R so $u_i(X_j) = u_i(\widetilde{X}_j)$.
2. If R is independent of $\widetilde{X}$, then $G_i(R \mid \widetilde{X}) = G_i(R)$, so the belief distribution no longer varies with $\widetilde{X}$. Under additive separability, $u_i(\widetilde{X}_j, R_j) = f_i(\widetilde{X}_j) + g_i(R_j)$, we have $\mathbb{E}_i[u_i(\widetilde{X}_j, R_j) \mid \widetilde{X}_j] = f_i(\widetilde{X}_j) + \mathbb{E}_i[g_i(R_j)] + \mathbb{E}_i[\epsilon_{ij} \mid \widetilde{X}_j]$. Since $\mathbb{E}_i[g_i(R_j)]$ does not vary with $\widetilde{X}_j$, it cancels in compensating differentials between any two jobs j and k , and no moments of $G_i(R)$ are identified. Identification of some belief moments under independence of R and $\widetilde{X}$ would require non-separable preferences. For example, if $u_i(\widetilde{X}_j, R_j) = f_i(\widetilde{X}_j) + g_i(R_j) + h_i(\widetilde{X}_j) \cdot R_j$, then $\mathbb{E}_i[u_i \mid \widetilde{X}_j] =$

¹For example in linearly separable models, $\mathbb{E}_i(R \mid \widetilde{X})$ is the moment of $G_i(R \mid \widetilde{X})$ that enters expected utility, and the belief parameters α_i are identified.

$f_i(\tilde{X}_j) + \mathbb{E}_i[g_i(R)] + h_i(\tilde{X}_j) \cdot \mathbb{E}_i[R] + \mathbb{E}_i[\epsilon_{ij} \mid \tilde{X}_j]$. Since $h_i(\cdot)$ is identified from complete scenarios as part of $u_i(\cdot)$, $\mathbb{E}_i[R]$ is identified from how compensating differentials in incomplete scenarios vary with $\tilde{X}_j$.

APPENDIX B: APPENDIX: NON-IDENTIFICATION USING CHOICE PROBABILITIES

Consider a set-up of complete and incomplete scenarios identical to ours, except where respondents are asked for choice probabilities, instead of making a choice and reporting compensating differentials for the other options. Utility of individual i from job j is given by $U_{ij} = X_j' \beta_i + \epsilon_{ij}$ where X_j consists of the exogenously varied relevant job attributes and β_i is the preference parameter vector of individual i . In the survey asking for choice probabilities, individuals are instructed to imagine themselves making a job choice at some point *in the future*, and correspondingly report probabilities today. Hence, the vector of error terms $\epsilon_i \equiv \{\epsilon_{i1}, \dots, \epsilon_{iJ}\}$ has the interpretation of resolvable uncertainty as in [Blass et al. \(2010\)](#). The usual assumption is that these unobserved resolvable uncertainties ϵ_i are independently and identically distributed across individuals following Type I extreme value distribution without loss of generality ([McFadden and Train, 2000](#)). Given the instructions, for each individual i , the additively separable unobserved resolvable uncertainties ϵ_i are independent of the exogenously varied X_j ([Wiswall and Zafar, 2018](#)).

Denote p_{ij} as the probability of individual i choosing job j in a complete scenario while observing all the attributes $X_j \equiv \{G_j, W_j, H_j, R_j\}$, and $\widetilde{p}_{ij}$ as the probability of choosing job j in an incomplete scenario while observing $\widetilde{X}_j \equiv X_j \setminus \{R_j\}$. With the assumption that the unobservable part of the utility function consists of resolvable uncertainty ϵ_{ij} i.i.d. following Type I extreme value distribution, we can write the probability of individual i choosing job j in the complete scenarios observing all the relevant attributes X_j as,

$$p_{ij} = \frac{\exp(X_j' \beta_i)}{\sum_j \exp(X_j' \beta_i)}$$

Note that in writing the choice probability like above requires the normalization of the variance of the error term as is standard in discrete choice models. However, in the incomplete scenarios individuals observe $\widetilde{X}_j$. Individuals do not observe R_j and use $R_j = \widetilde{X}_j' \alpha_i^x + \eta_j$ to form their beliefs where η_j could be interpreted as measurement error. In the incomplete scenarios, the probability of individual i choosing job j over job k for all jobs $j \neq k$ can be written as:

$$\begin{aligned} \widetilde{p}_{ij} &= \Pr(U_{ij} > U_{ik}) \\ &= \Pr \left(\sum_{x \in \widetilde{X}} \beta_i^x x_j + \beta_i^R R_j + \epsilon_{ij} > \sum_{x \in \widetilde{X}} \beta_i^x x_k + \beta_i^R R_k + \epsilon_{ik} \right) \\ &= \Pr \left(\sum_{x \in \widetilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) x_j + \beta_i^R \eta_j + \epsilon_{ij} > \sum_{x \in \widetilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) x_k + \beta_i^R \eta_k + \epsilon_{ik} \right) \end{aligned}$$

$$= Pr \left(\beta_i^R (\eta_j - \eta_k) + \epsilon_{ij} - \epsilon_{ik} > \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) (x_k - x_j) \right)$$

Observe that calculating this choice probability requires the researcher to normalize of the variance of the error term in this incomplete scenario which is $\beta_i^R \eta_j + \epsilon_{ij}$, on top of the above normalization of the variance of ϵ_{ij} . The requirement of having to normalize two variances makes one of the normalizations non-innocuous. This is the primary disadvantage of using choice probabilities in contexts such as ours where the individuals make choices in both complete and incomplete scenarios.

B.1 Non-identification using choice probabilities with additional parametric assumptions

It is important to highlight that even parametric assumption on the distribution of η will not achieve identification. For the purpose of illustration and simplicity, let us assume that both error terms follow normal distributions.² In particular, let us assume that $\epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma_\epsilon^2)$ and $\eta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\eta^2)$ with $\epsilon_i \perp \eta_j$ for all i and j . Given this we have,

$$\beta_i^R \eta_j + \epsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma_\epsilon^2 + (\beta_i^R \sigma_\eta)^2)$$

With choice probabilities reported in the complete scenarios it is easy to show that as number of complete and incomplete scenarios go to infinity one can identify for all individuals i the following parameters from the complete scenarios

$$\left\{ \frac{\beta_i^x}{\sigma_\epsilon} \right\} \quad \forall x \in X \equiv \{G, W, H, R\}$$

and for all individuals i , the following parameters from the incomplete scenarios

$$\left\{ \frac{\beta_i^x + \beta_i^R \alpha_i^x}{\sqrt{\sigma_\epsilon^2 + (\beta_i^R \sigma_\eta)^2}} \right\} \quad \forall x \in \tilde{X} \equiv \{G, W, H\}$$

From the above it is easy to see that the belief parameters α_i^x are not identified without non-innocuous normalizations of the variance terms. Even with the distributional assumption on η the only way to achieve identification of the belief parameters will require the knowledge of σ_η . To see this observe that for all $x \in \tilde{X} \equiv \{G, W, H\}$ we can rewrite $\left\{ \frac{\beta_i^x + \beta_i^R \alpha_i^x}{\sqrt{\sigma_\epsilon^2 + (\beta_i^R \sigma_\eta)^2}} \right\}$ as

²If we proceeded with the assumption of ϵ_i following Type-I extreme value, for any distribution of η_j the distribution of $\beta_i^R \eta_j + \epsilon_{ij}$ obtained by convolution would no longer follow a Type I extreme value distribution. Consequently, the differences would no longer follow a logistic distribution, and we will lose the convenience of the closed form solution which is helpful for the purposes of illustration.

$$\left\{ \frac{\frac{\beta_i^x}{\sigma_\epsilon} + \frac{\beta_i^R}{\sigma_\epsilon} \alpha_i^x}{\sqrt{1 + \left(\frac{\beta_i^R}{\sigma_\epsilon} \sigma_\eta \right)^2}} \right\} \forall x \in \tilde{X} \equiv \{G, W, H\}$$

With $\left\{ \frac{\beta_i^G}{\sigma_\epsilon}, \frac{\beta_i^W}{\sigma_\epsilon}, \frac{\beta_i^H}{\sigma_\epsilon}, \frac{\beta_i^R}{\sigma_\epsilon} \right\}$ identified for all individuals i from the complete scenarios, the only unknown terms in the above expression for all individuals i are the set of belief parameters $\{\alpha_i^G, \alpha_i^W, \alpha_i^H\}$ and σ_η .

APPENDIX C: APPENDIX: COMPARISON WITH [ANDREW AND ADAMS \(2025\)](#)

Both our paper and [Andrew and Adams \(2025\)](#) (henceforth AA) use similar intuition to separate beliefs from preferences, but fundamentally differ in the underlying primitives and the assumptions that achieve identification. Specifically, both papers differ in the type of beliefs, primarily based on what a decision-maker observes and what remains unresolved at the point of decision-making, and thus are applicable to different classes of decision-making problems.

The method of AA applies to a class of optimal stopping problems where agents make a sequence of decisions over time (e.g., stop/continue). As in standard discrete choice models, agents draw idiosyncratic preference shocks in each period from a known distribution. Additionally, the agent does not have perfect information about all *future* realizations of a two-dimensional discrete stochastic state variable, for e.g., the quality of a marriage offer $q_t \in \{L, H\}$ in their application. Importantly, the agent only observes q_t once period t arrives. The beliefs in AA are the probabilities the agent places on each realization of q_t in future periods. Conditional on the current realization and beliefs over future realizations, the agent decides at each period whether to stop and obtain the terminal payoff, or continue. This makes any revealed preference stopping decisions only informative about a mix of preferences and beliefs about future q_t which motivates their survey design, similar in intuition to ours.

First, these beliefs in AA are fundamentally different from the beliefs our paper addresses. In AA, uncertainty is purely about future realizations because each element of the discrete state space is observed perfectly once the relevant period arrives. In contrast, our setting deals with situations where the state space in the current period is not perfectly observed. For example, in our setting a worker does not perfectly observe manager quality while choosing between multiple heterogeneous job offers, while in AA groom quality at t is perfectly observed at t . Our paper thus addresses situations where agents compare multiple options, each characterized by its own unobserved (continuous or discrete) quality, and thus need to form expectations about these unobserved attributes conditional on observed attributes to maximize expected utility. This makes identification in AA rely on the agent perfectly observing the realization of q_t at t . In dynamic problems as in AA, if q_t were imperfectly observed at t , the ex post experiment would no longer identify preferences. Thus, comparing choice probabilities across different information structures would face the same normalization problem we formalize in Appendix B that will lead to non-identification of beliefs.

Second, the differences in the structure of beliefs also imposes different requirements on identifying variation. In AA, when the dimension of q_t is more than two, identification of beliefs additionally requires preference-shifting instrumental variables that affect terminal payoffs but are excluded from beliefs (See Appendix B.4 in AA).³ In our

³This is not a limitation, but rather a requirement that arises from the structure of optimal stopping problems. Since the continuation value is a probability weighted average of state-contingent values, when $\dim(q) = k > 2$, identifying its $k - 1$ probability weights from this weighted average requires variation that shifts the values while holding probabilities fixed.

framework, richer belief functions only require additional scenarios but no instrumental variables. As shown in our identification proofs, as the number of scenarios grows, our belief function is non-parametrically identified.

Finally, the differences in the survey instruments of the two papers reflect differences of the primitives. Parallely, our survey design situates respondents in resolving current informational uncertainty,⁴ while the survey design of AA situate respondents in resolving future uncertainty.

A similarity on non-identification in both methods arise from whether individuals care about the unobserved attribute (in our case) or the unobserved future state (in AA). If individuals are indifferent to any variation in the unobserved attribute (in our case) or the unobserved future state (in AA), then beliefs are fundamentally unidentified by construction in both methods because any belief distribution would be consistent with observed choices.

The two approaches are thus complements, determined by the decision problem's structure. Our method applies where agents compare multiple options each with its own draw of an unobserved attribute. Examples of applications include school choice with unobserved school quality, physician choice with unobserved provider quality, housing choice with unobserved neighborhood quality, firm hiring with unobserved worker productivity, etc. Respectively, researchers could study parental beliefs about school quality by varying school performance metrics and eliciting tuition compensating differentials, beliefs about physician quality by varying provider quality information and eliciting fee differentials, beliefs about neighborhood quality by varying neighborhood information and eliciting willingness to pay for housing/rent differentials, and employer beliefs about worker quality by varying performance information and eliciting wage offers. In contrast, AA address settings where agents perfectly observe the current realization of a discrete state variable but hold beliefs about the probability distribution of its two-dimensional future realizations, and must decide whether to stop and obtain the terminal payoff or continue. Examples of applications include job search where workers hold beliefs about the probability of receiving high/low future wage offers, fertility decisions where parents hold beliefs about the probability of a future conception, and retirement decisions where workers hold beliefs about the probability of future good/bad health shock. The crucial requirement for identification with choice probabilities in AA is that the unobserved variable is perfectly observed at each point in time. If the researcher's context is such that this requirement is violated, the normalization problem we highlight would apply.

Existing structural models typically assume either rational expectations or estimate these from population-level data. But not only can subjective beliefs be very different from rational expectations (see for e.g., [Kapor et al. \(2020\)](#), [Jäger et al. \(2021b\)](#)) but also estimates from population-level data will mask policy relevant heterogeneity in beliefs. Thus, regardless of context, identification of subjective beliefs (and preferences) is crucial. Together, our paper and AA provide researchers with complementary identification

⁴We specifically instruct respondents to assume the three jobs in each scenario have been offered to them (see Appendix Figure F.1).

strategies for two fundamental forms of uncertainty that arise in structural models of choice. Our paper deals with beliefs arising from uncertainty about attributes of currently available options while AA deal with beliefs arising from uncertainty about future states of the world. Both our paper and AA highlight the idea that appropriately designed information experiments by varying what respondents observe across hypothetical scenarios provides researchers with a flexible tool for identifying beliefs.

APPENDIX D: APPENDIX: ESTIMATION DETAILS

The compensating differentials are reported with two different but independent measurement errors. First, respondents report in multiples of five (Figure 2). This is also common in surveys asking for choice probabilities (Blass et al., 2010). Second, the slider interface introduces random error despite displaying exact amounts. Under classical measurement error assumptions on both, these affect only standard errors, not consistency.

Denote as Δ_{ijk}^* and $\tilde{\Delta}_{ijk}^*$ the latent compensating differentials and as e_i and $\tilde{e}_i$ the composite classical measurement errors in the complete and incomplete scenarios, respectively. Thus, we have the following set of estimating equations for each individual i and each pair of jobs j and k within every scenario:

$$\begin{aligned}\Delta_{ijk}^* &= \sum_{x \in X} \beta_i^x (x_j - x_k) + e_i \\ \tilde{\Delta}_{ijk}^* &= \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) (x_j - x_k) + \tilde{e}_i\end{aligned}\tag{D.1}$$

Under classical measurement error we obtain the following moment conditions:

$$\begin{aligned}\mathbb{E}[\Delta_{ijk}^* \mid X_j, X_k] &= \sum_{x \in X} \beta_i^x (x_j - x_k) = (X_j - X_k)' \beta_i \\ \mathbb{E}[\tilde{\Delta}_{ijk}^* \mid \tilde{X}_j, \tilde{X}_k] &= \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) (x_j - x_k) = (\tilde{X}_j - \tilde{X}_k)' \tilde{\beta}_i\end{aligned}\tag{D.2}$$

We jointly estimate system (D.1) using constrained least squares normalizing $\beta_i^W = 1$. The equations have no constant by construction since they are derived from utility differences. We use the block bootstrap at the respondent level allowing for arbitrary correlation among responses within each respondent. We stack up the matrices of observables across the complete and incomplete scenarios, and estimate the following constrained least squares regression with no constant:

$$\begin{bmatrix} \Delta_{jk} \\ \vdots \\ \tilde{\Delta}_{jk} \\ \vdots \end{bmatrix} = \begin{bmatrix} \mathbf{X}_j - \mathbf{X}_k & \mathbf{0} \\ \vdots & \vdots \\ \mathbf{0} & \tilde{\mathbf{X}}_j - \tilde{\mathbf{X}}_k \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} \beta \\ \tilde{\beta} \end{bmatrix} + \mathbf{e}\tag{D.3}$$

where the constraint is the normalization for the preference parameter on wages to be equal to one. The standard errors are computed using the block bootstrap at the student level. This accounts for any arbitrary correlation between responses at the student level.

The block bootstrap algorithm is as follows: The sample contains N individuals.

1. Generate B the block bootstrap samples of N individuals each.

For each $b = 1, \dots, B$

2. Estimate the model for each member in the bootstrap sample by bootstrapping each member's responses.

Obtain $\hat{\beta}_i^{(b)}$ and compute its sample mean $\hat{\beta}^{(b)} = \sum_{i=1}^N \hat{\beta}_i^{(b)}$.

Compute the mean and standard deviation of the B estimates in hand to generate estimates of the bootstrap mean and bootstrap standard error.

Preferences:

$$\hat{\mathbb{E}}(\beta_i) = \frac{1}{B} \sum_{b=1}^B \hat{\beta}^{(b)}$$

$$\widehat{\text{std error}}(\beta_i) = SD(\hat{\beta}^{(b)})$$

Preferences confounded with valuation of beliefs:

$$\hat{\mathbb{E}}(\tilde{\beta}_i) = \frac{1}{B} \sum_{b=1}^B \hat{\beta}^{(b)}$$

$$\widehat{\text{std error}}(\tilde{\beta}_i) = SD(\hat{\beta}^{(b)})$$

Valuation of beliefs:

$$\hat{\mathbb{E}}(\tilde{\beta}_i - \beta_i) = \frac{1}{B} \sum_{b=1}^B \widehat{\tilde{\beta} - \beta}^{(b)}$$

$$\widehat{\text{std error}}(\tilde{\beta}_i - \beta_i) = SD(\widehat{\tilde{\beta} - \beta}^{(b)})$$

FIGURE D.1. Bootstrap distribution of beliefs about male managers' mentorship and preferences to work for male managers

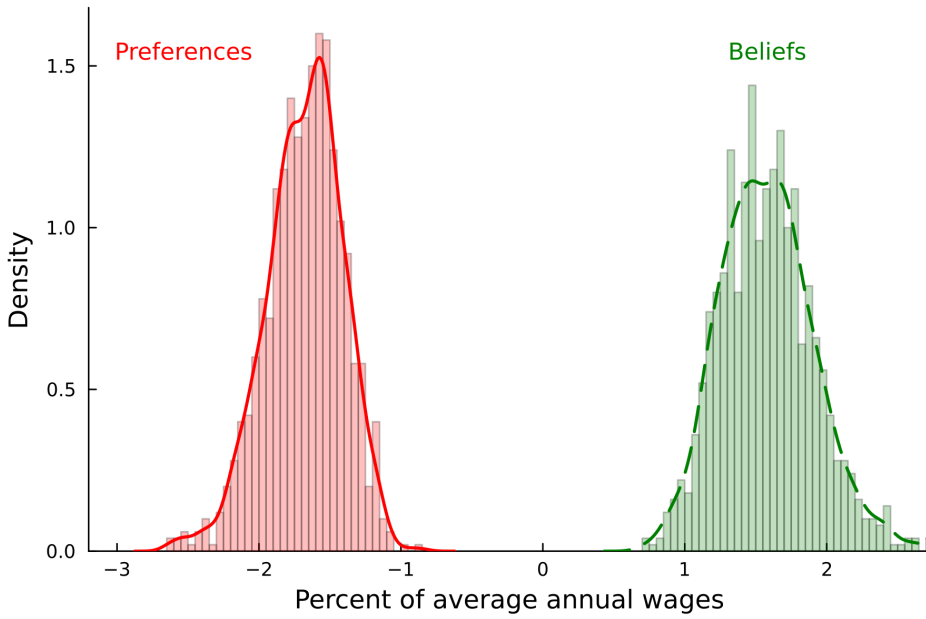


FIGURE D.1. *Notes:* The figure shows bootstrap distributions of beliefs about male manager mentorship and preferences to work for male managers, relative to female managers in the percentage of average annual wages. The bootstrap distributions are obtained from 1,000 block bootstrapped samples, using the algorithm described in Appendix D. These bootstrap distributions are used for estimation of means and standard errors of preferences and beliefs.

APPENDIX E: APPENDIX: INCORPORATING OTHER MANAGERIAL QUALITIES IN THE MODEL

Let $i = 1, \dots, N$ index individuals. Each individual i , for each managerial quality (other than mentorship) indexed by $q = 1, \dots, Q$, provides a single ordinal statement $Y_{iq} \in \{-1, 0, +1\}$ about whether they perceive a male manager to be better, similar, or worse than a female manager in the q^{th} quality Q_q . Hence, for individual i , the observed perception data are $\{(Y_{iq})_{q=1, \dots, Q}\} \in \{-1, 0, +1\}^Q$.

A consequent question that arises is whether the negative beliefs about female manager mentorship that we document are robust conditional on other managerial qualities. To this end, the parameter of interest can be represented as the difference in expected mentorship quality between female and male managers conditional on other managerial qualities $\mathbb{Q} \equiv \{Q_1, \dots, Q_K\} \in \mathbb{R}^K$ while suppressing other job attributes to reduce notational clutter:

$$E[R_j \mid G_j = F, \mathbb{Q}] - E[R_j \mid G_j = M, \mathbb{Q}] \quad (\text{E.1})$$

E.1 Utility function

The utility of individual i from job j is given by:

$$U_{ij} = X_j' \beta_i + \sum_{q=1}^Q \beta_i^q Q_{iq}^{g(j)} + \epsilon_{ij} \quad (\text{E.2})$$

where $X_j \equiv (G_j, W_j, R_j, H_j)'$ is the vector of observable attributes of job j , $Q_{iq}^{g(j)}$ is the perceived latent quality of the manager of job j for individual i . The preference parameters of individual i is given by the vector:

$$\beta_i \cup \{\beta_i^q\}_{q=1}^Q \equiv (\beta_i^G, \beta_i^W, \beta_i^H, \beta_i^R)' \cup \left(\{\beta_i^q\}_{q=1}^Q \right)' \in \mathbb{R}^{K+1+Q}$$

E.2 Latent Quality

Let individual i 's perception of the q^{th} latent quality of manager of gender $g \in \{M, F\}$ be denoted by Q_{iq}^g . We model the latent qualities Q_{iq}^g as:

$$Q_{iq}^g = \mu_q + \delta_q \mathbb{1}(g = M) + \nu_{iq}^g \quad \nu_{iq}^g \sim \mathcal{N}(0, \sigma_q^{g^2}), \quad (\text{E.3})$$

where, δ_q captures average beliefs that male managers exceed female managers in quality q , $(\nu_{iq}^M, \nu_{iq}^F)' \sim \mathcal{N}(0, \Sigma_q)$ and mean independent of other parameters and variables, Σ_q is the covariance matrix capturing the correlations between the errors across different qualities given by:

$$\Sigma_q = \begin{pmatrix} \sigma_q^{M^2} & \rho_k \sigma_q^{MF} \\ \rho_k \sigma_q^{MF} & \sigma_q^{F^2} \end{pmatrix}$$

Denote $D_{iq} \equiv Q_{iq}^M - Q_{iq}^F$, the difference in i 's perceptions of the q^{th} quality between male and female managers, and $\eta_{iq} \equiv (\nu_{iq}^M - \nu_{iq}^F)$. Then given the parametrization of the latent qualities we have:

$$D_{iq} \equiv Q_{iq}^M - Q_{iq}^F = \delta_q + \eta_{iq} \quad (\text{E.4})$$

Thus, for each individual i and quality q , we can represent their perception data as:

$$Y_{iq} = \begin{cases} +1 & \text{if } D_{iq} > \overline{\tau}_q, \\ 0 & \text{if } D_{iq} \in [-\underline{\tau}_q, \overline{\tau}_q], \\ -1 & \text{if } D_{iq} < -\underline{\tau}_q \end{cases} \quad (\text{E.5})$$

where $\overline{\tau}_q$ and $\underline{\tau}_q$ are thresholds that partition the q^{th} continuous latent quality difference into three categories observed in the data. Given distributional assumptions, we have $D_{iq} \sim \mathcal{N}(\delta_q, \sigma_{D_q}^2)$, where $\sigma_{D_q}^2 \equiv \sigma_q^{M^2} + \sigma_q^{F^2} - 2\rho_q \sigma_q^{MF}$.

E.3 Identification

Given the setup and distributional assumptions this leads us to an ordered probit model. The shares of individuals with perceptions $Y_{iq} = +1, 0, -1$ can be expressed as:

$$\begin{aligned} P(Y_{iq} = +1) &= P(D_{iq} > \overline{\tau}_q) = 1 - \Phi\left(\frac{\overline{\tau}_q - \delta_q}{\sigma_{D_q}}\right) \\ P(Y_{iq} = 0) &= P(-\underline{\tau}_q \leq D_{iq} \leq \overline{\tau}_q) = \Phi\left(\frac{\overline{\tau}_q - \delta_q}{\sigma_{D_q}}\right) - \Phi\left(\frac{-\underline{\tau}_q - \delta_q}{\sigma_{D_q}}\right) \\ P(Y_{iq} = -1) &= P(D_{iq} < -\underline{\tau}_q) = \Phi\left(\frac{-\underline{\tau}_q - \delta_q}{\sigma_{D_q}}\right) \end{aligned}$$

We have a system of two linearly independent equations with four parameters $\Theta_q \equiv \{-\delta_q, \sigma_{D_q}, \overline{\tau}_q, \underline{\tau}_q\}$ for each quality q . We normalize one of the thresholds, $\underline{\tau}_q$, to zero and the variance $\sigma_{D_q}^2$ to 1. This gives us two equations in two unknowns for each quality q :

$$\begin{aligned} P(Y_{iq} = +1) &= 1 - \Phi(\overline{\tau}_q - \delta_q) \\ P(Y_{iq} = -1) &= \Phi(-\delta_q) \end{aligned}$$

Consequently, the empirical shares of perceptions point identify the parameters $\{\delta_q, \overline{\tau}_q\}$ for each quality q .

E.3.1 Complete scenarios In the complete scenarios, individuals observe all attributes in set X for each job. The expected utility of individual i from job j conditional on its observable attributes and their perceptions in the complete scenarios is given by

$$\mathbb{E}_i[U_{ij} | X_j, \mathbb{Q}] = X_j' \beta_i + \sum_{q=1}^Q \beta_i^q Q_{iq}^{g(j)} + \mathbb{E}_i(\epsilon_{ij} | X_j)$$

Each individual i knows their preferences and perceptions and hence do not take expectations over them. Note that given our model of latent qualities, we can express $Q_{iq}^{g(j)}$ as:

$$\begin{aligned} Q_{iq}^{g(j)} &= \mu_q + \delta_q \mathbb{1}(g(j) = M) + \nu_{iq}^{g(j)} \\ &= \mu_q + \delta_q G_j + \nu_{iq}^{g(j)} \\ &= \mu_q + \delta_q G_j + \nu_{iq}^M G_j + \nu_{iq}^F (1 - G_j) \\ &= \mu_q + (\delta_q + \eta_{iq}) G_j + \nu_{iq}^F \end{aligned}$$

Substituting $Q_{iq}^{g(j)} = \mu_q + (\delta_q + \eta_{iq}) G_j + \nu_{iq}^F$, we can express the expected utility as:

$$\begin{aligned} \mathbb{E}_i [U_{ij} | X_j, \mathbb{Q}] &= X_j' \beta_i + \sum_{q=1}^Q \beta_i^q \left(\mu_q + (\delta_q + \eta_{iq}) G_j + \nu_{iq}^F \right) + \mathbb{E}_i (\epsilon_{ij} | X_j) \\ &= X_j' \beta_i + \sum_{q=1}^Q \beta_i^q \mu_q + \sum_{q=1}^Q \beta_i^q (\delta_q + \eta_{iq}) G_j + \sum_{q=1}^Q \beta_i^q \nu_{iq}^F + \mathbb{E}_i (\epsilon_{ij} | X_j) \\ &= \theta_i + X_j' \beta_i + \left(\sum_{q=1}^Q \beta_i^q (\delta_q + \eta_{iq}) \right) G_j + \mathbb{E}_i (\epsilon_{ij} | X_j) \end{aligned}$$

where $\theta_i \equiv \sum_{q=1}^Q \beta_i^q \mu_q + \sum_{q=1}^Q \beta_i^q \nu_{iq}^F$ is a constant term for individual i . Expanding $X_j = (G_j, W_j, R_j, H_j)'$, we can express the expected utility as:

$$\begin{aligned} \mathbb{E}_i [U_{ij} | X_j, \mathbb{Q}] &= \theta_i + \beta_i^G G_j + \beta_i^W W_j + \beta_i^R R_j + \beta_i^H H_j \\ &\quad + \left(\sum_{q=1}^Q \beta_i^q (\delta_q + \eta_{iq}) \right) G_j + \mathbb{E}_i (\epsilon_{ij} | X_j) \\ &= \theta_i + \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q (\delta_q + \eta_{iq}) \right) G_j \\ &\quad + \beta_i^W W_j + \beta_i^R R_j + \beta_i^H H_j + \mathbb{E}_i (\epsilon_{ij} | X_j) \\ &= \theta_i + \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q \right) G_j + \sum_{q=1}^Q \beta_i^q \eta_{iq} G_j \\ &\quad + \beta_i^W W_j + \beta_i^R R_j + \beta_i^H H_j + \mathbb{E}_i (\epsilon_{ij} | X_j) \end{aligned}$$

Note that δ_q is identified from the perception data as described in the previous section. However, η_{iq} is unknown. We estimate η_{iq} for each i and for all q with its expectation conditional on individual i 's perception data of the q^{th} quality. Using properties of truncated normal distributions, for each quality q we have:

$$\widehat{\eta}_{iq} = E[\eta_{iq} | Y_{iq}] = \begin{cases} E[\eta_{iq} | Y_{iq} = +1] = E[\eta_{iq} | \eta_{iq} > \bar{\tau}_q - \delta_q] & = \frac{\phi(\widehat{\tau}_q - \widehat{\delta}_q)}{1 - \Phi(\widehat{\tau}_q - \widehat{\delta}_q)} \\ E[\eta_{iq} | Y_{iq} = 0] = E[\eta_{iq} | -\delta_q \leq \eta_{iq} \leq \bar{\tau}_q - \delta_q] & = \frac{\phi(-\widehat{\delta}_q) - \phi(\widehat{\tau}_q - \widehat{\delta}_q)}{\Phi(\widehat{\tau}_q - \widehat{\delta}_q) - \Phi(-\widehat{\delta}_q)} \\ E[\eta_{iq} | Y_{iq} = -1] = E[\eta_{iq} | \eta_{iq} < -\delta_q] & = -\frac{\phi(-\widehat{\delta}_q)}{\Phi(-\widehat{\delta}_q)} \end{cases}$$

where $\widehat{\delta}_q$ and $\widehat{\tau}_q$ are estimated from the first-step ordered probit model for latent managerial qualities.

Thus expressing the expected utility in terms of the estimated η_{iq} , we have:

$$\mathbb{E}_i [U_{ij} | X_j, \mathbb{Q}] = \theta_i + \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q \right) G_j + \sum_{q=1}^Q \beta_i^q \widehat{\eta}_{iq} G_j + \beta_i^W W_j + \beta_i^R R_j + \beta_i^H H_j + \mathbb{E}_i (\epsilon_{ij} | X_j)$$

Maintaining assumptions (1) and (2) described in section 5, the difference in expected utility between jobs j and k for individual i given the reported compensating differentials Δ_{ijk} is given by:

$$\begin{aligned} \Delta_{ijk} = & \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q \right) (G_j - G_k) + \sum_{q=1}^Q \beta_i^q [\widehat{\eta}_{iq} (G_j - G_k)] \\ & + \beta_i^W (W_j - W_k) + \beta_i^R (R_j - R_k) + \beta_i^H (H_j - H_k) \end{aligned} \quad (\text{E.6})$$

This identifies the parameters: $\left\{ \left\{ \beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q \right\}_{q=1}^Q, \left\{ \beta_i^q \right\}_{q=1}^Q, \beta_i^R, \beta_i^H \right\}$ for each individual i . Given that $\{\delta_q\}_{q=1}^Q$ is identified from the perception data, we can separately identify the parameters $\{\beta_i^G\}$ for each individual i .

E.3.2 Incomplete scenarios In the incomplete scenarios, individuals do not observe R_j and hence take expectations over it while computing their expected utility from job j in the incomplete scenarios. We model the mentorship quality R_j in terms of the attributes of job j and individual i 's perceived latent qualities $Q_{iq}^{g(j)}$ of the manager of gender $g(j) \in \{M, F\}$ as follows (suppressing other job attributes to reduce notational clutter):

$$R_j = \alpha_i^G G_j + \sum_{q=1}^Q \alpha_i^q Q_{iq}^{g(j)} + u_{ij} \quad (\text{E.7})$$

where $E \left[u_{ij} | G_j, \left\{ Q_{iq}^{g(j)} \right\}_q \right] = 0$ for all i, j, q .

Replacing this in the expression for R_j , we have:

$$\begin{aligned}
 R_j &= \alpha_i^G G_j + \sum_{q=1}^Q \alpha_i^q \left(\mu_q + (\delta_q + \eta_{iq}) G_j + \nu_{iq}^F \right) + u_{ij} \\
 &= \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq}) \right) G_j + \sum_{q=1}^Q \alpha_i^q \mu_q + \sum_{q=1}^Q \alpha_i^q \nu_{iq}^F + u_{ij} \\
 &= \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq}) \right) G_j + C_i + u_{ij}
 \end{aligned}$$

where $C_i \equiv \sum_{q=1}^Q \alpha_i^q \mu_q + \sum_{q=1}^Q \alpha_i^q \nu_{iq}^F$ is constant for individual i . Subsequently,

$$\begin{aligned}
 E_i[R_j \mid G_j, \mathbb{Q}] &= E_i \left[R_j \mid G_j, \left\{ Q_{iq}^{g(j)} \right\}_q \right] \\
 &= \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq}) \right) G_j + C_i
 \end{aligned}$$

Thus our parameter of interest can be expressed as:

$$E_i[R_j \mid G_j = M, \mathbb{Q}] - E_i[R_j \mid G_j = F, \mathbb{Q}] = \alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq}) \quad (\text{E.8})$$

Expanding this to include other job attributes W_j and H_j , we have:

$$E_i[R_j \mid \widetilde{X}_j, \mathbb{Q}] = \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq}) \right) G_j + C_i + \alpha_i^W W_j + \alpha_i^H H_j$$

where $\widetilde{X}_j \equiv (G_j, W_j, H_j)'$ is the vector of observable attributes in the incomplete scenarios.

The expected utility of individual i from job j in the incomplete scenarios is given by:

$$\begin{aligned}
 \mathbb{E}_i \left[U_{ij} \mid \widetilde{X}_j, \mathbb{Q} \right] &= \theta_i + \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q \right) G_j \\
 &\quad + \sum_{q=1}^Q \beta_i^q \widehat{\eta}_{iq} G_j + \beta_i^W W_j + \beta_i^H H_j \\
 &\quad + \beta_i^R \mathbb{E}_i[R_j \mid \widetilde{X}_j, \mathbb{Q}] + \mathbb{E}_i \left(\epsilon_{ij} \mid \widetilde{X}_j, \mathbb{Q} \right)
 \end{aligned}$$

Substituting the expression for $E_i[R_j | \widetilde{X}_j, \mathbb{Q}]$, we have:

$$\begin{aligned}
 \mathbb{E}_i [U_{ij} | \widetilde{X}_j, \mathbb{Q}] &= \theta_i + \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q \right) G_j \\
 &\quad + \sum_{q=1}^Q \beta_i^q \widehat{\eta}_{iq} G_j + \beta_i^W W_j + \beta_i^H H_j \\
 &\quad + \beta_i^R \left(\left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \widehat{\eta}_{iq}) \right) G_j + C_i + \alpha_i^W W_j + \alpha_i^H H_j \right) \\
 &\quad + \mathbb{E}_i (\epsilon_{ij} | \widetilde{X}_j, \mathbb{Q}) \\
 &= \theta_i + \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q \right) G_j + \sum_{q=1}^Q \beta_i^q \widehat{\eta}_{iq} G_j \\
 &\quad + \beta_i^W W_j + \beta_i^H H_j \\
 &\quad + \beta_i^R \left(\left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q \right) G_j + \sum_{q=1}^Q \alpha_i^q \widehat{\eta}_{iq} G_j + C_i + \alpha_i^W W_j + \alpha_i^H H_j \right) \\
 &\quad + \mathbb{E}_i (\epsilon_{ij} | \widetilde{X}_j, \mathbb{Q})
 \end{aligned}$$

This can be simplified to:

$$\begin{aligned}
 \mathbb{E}_i [U_{ij} | \widetilde{X}_j, \mathbb{Q}] &= \theta_i + \beta_i^R C_i \\
 &\quad + \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q + \beta_i^R \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q \right) \right) G_j \\
 &\quad + \sum_{q=1}^Q (\beta_i^q + \beta_i^R \alpha_i^q) \widehat{\eta}_{iq} G_j \\
 &\quad + (\beta_i^W + \beta_i^R \alpha_i^W) W_j + (\beta_i^H + \beta_i^R \alpha_i^H) H_j \\
 &\quad + \mathbb{E}_i (\epsilon_{ij} | \widetilde{X}_j, \mathbb{Q})
 \end{aligned}$$

Using reported compensating differentials $\widetilde{\Delta}_{ijk}$, and imposing assumptions (1) and (2) now conditional on perceptions, we can express the difference in expected utility between jobs j and k for individual i in the incomplete scenarios as:

$$\widetilde{\Delta}_{ijk} = \left(\beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q + \beta_i^R \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q \right) \right) (G_j - G_k)$$

$$\begin{aligned}
& + \sum_{q=1}^Q (\beta_i^q + \beta_i^R \alpha_i^q) \widehat{\eta}_{iq} (G_j - G_k) \\
& + (\beta_i^W + \beta_i^R \alpha_i^W) (W_j - W_k) + (\beta_i^H + \beta_i^R \alpha_i^H) (H_j - H_k)
\end{aligned} \tag{E.9}$$

This for each individual i , identifies the parameters:

$$\left\{ \left\{ \beta_i^G + \sum_{q=1}^Q \beta_i^q \delta_q + \beta_i^R \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q \right) \right\}_{q=1}^Q, \right. \\
\left. \left\{ \beta_i^q + \beta_i^R \alpha_i^q \right\}_{q=1}^Q, \beta_i^W + \beta_i^R \alpha_i^W, \beta_i^H + \beta_i^R \alpha_i^H \right\}$$

Given that $\{\delta_q\}_{q=1}^Q$ is identified from the perception data, and the variation in compensating differentials in the complete scenarios identify the parameters $\{\beta_i^G, \beta_i^H, \beta_i^R, \{\beta_i^q\}_{q=1}^Q\}$ for each individual i , the variation between the incomplete and the complete scenarios separately identify the parameters $\{\alpha_i^G, \alpha_i^W, \alpha_i^H, \alpha_i^R, \{\alpha_i^q\}_{q=1}^Q\}$ for each individual i .

Thus our parameter of interest of the beliefs about the gender gap in mentorship conditional on other managerial qualities $\mathbb{Q}$ is identified by:

$$E_i[R_j \mid G_j = F, \mathbb{Q}] - E_i[R_j \mid G_j = M, \mathbb{Q}] = \alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq})$$

The valuation of this parameter of interest is simply:

$$\beta_i^R \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq}) \right)$$

Further, it is important to note that the average of the individual-specific parameter of interest across all individuals i is equal to

$$\begin{aligned}
& \frac{1}{N} \sum_{i=1}^N \beta_i^R \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q (\delta_q + \eta_{iq}) \right) \\
& = \frac{1}{N} \sum_{i=1}^N \beta_i^R \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q \right) + \frac{1}{N} \sum_{i=1}^N \beta_i^R \left(\sum_{q=1}^Q \alpha_i^q \eta_{iq} \right)
\end{aligned}$$

Note that the second term $\frac{1}{N} \sum_{i=1}^N \left(\sum_{q=1}^Q \alpha_i^q \eta_{iq} \right)$ goes to zero as $N \rightarrow \infty$, since η_{iq} is the difference in the latent qualities and follows a symmetric distribution around zero and mean independent of other parameters and variables. Thus, the average of the parameter of interest across all individuals i boils down to:

$$\beta_i^R \left(\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q \right)$$

The beliefs about the gender gap in mentorship conditional on other managerial qualities can be decomposed into two components. The first component is trait-driven, $\frac{\beta_i^R \sum_{q=1}^Q \alpha_i^q \delta_q}{\beta_i^R (\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q)}$ reflecting the share of beliefs about the gender gap in mentorship driven by perceived gender gap in traits. The second component reflects the residual share in beliefs conditional on gender, $\frac{\beta_i^R \alpha_i^G}{\beta_i^R (\alpha_i^G + \sum_{q=1}^Q \alpha_i^q \delta_q)}$ reflecting respondents' belief about mentorship differences based on gender other than the considered traits.

E.4 Estimation and Inference

We implement a block-bootstrap procedure with 1,000 replications that resamples individuals and re-estimates all of the three following steps:

Step 1: We estimate gender differences in perceived managerial qualities $\{\delta_q\}$ using ordered probit models on the perception data $\{Y_{iq}\}$.

Step 2: Using the first-stage estimates, we compute the residuals $\widehat{\eta}_{iq} = E[\eta_{iq}|Y_{iq}]$ using properties of truncated normal distributions.

Step 3: We jointly estimate individual-specific preference and belief parameters by stacking the compensating differential equations from both complete (equation E.6) and incomplete scenarios (equation E.9) into a single system. We estimate this system via constrained least squares with $\beta_i^W = 1$ normalized similar to our main estimation.

Similar to our original estimation and inference approach detailed in Appendix Section D, we use the bootstrap distribution for both point estimates and inference. Our reported parameter estimates and their standard errors are the means and standard deviations of the bootstrap distribution respectively. This approach accounts for the multi-step estimation error, generated regressor problem and arbitrary within-individual correlation.

E.5 Results

We summarize the main results from incorporating other perceived managerial traits into the model.

Panel A of Table G shows that after accounting for beliefs about latent traits, workers still exhibit a preference for female managers in the complete scenarios ($\beta_i^G + \sum_q \beta_i^q \delta_q = -0.045$, significant at 1% level). Given estimates of $\beta_i^q \delta_q$ for all q we can estimate $\beta_i^G = -0.037$ (0.52% of wages) as the pure gender preferences. This magnitude in comparison to our baseline model ($\beta_i^{G(baseline)} = -0.119$) is lower by more than half. This result provides empirical support for the idea that once we make men and women more similar along more dimensions, reassuringly the "pure" preference for gender declines substantially. We discuss this more in the main text of our paper.

Panel B shows that the strongest perceived trait difference between male and female managers is in competence ($\delta_{q1} = 0.96$), and this trait is also significantly weighted in how individuals infer mentorship ($\beta_i^R \alpha_i^{q1} = 0.015$, significant at 10% level).

Most importantly, Panel C reports the beliefs about the gender gap in mentorship conditional on other traits and how much of it is driven by perceived gender gap in other traits. The total valuation of beliefs about the gender gap in mentorship once conditioned on other traits reduces to 0.6% of wages. Compared to our baseline estimates this represents a reduction of 62.5%. Our model provides a useful decomposition of this conditional beliefs about the gender gap in mentorship. Of this, 43.6% is attributable to perception of gender gaps in other traits. Thus, a substantial portion of the beliefs about the gender gap in mentorship, reflects how individuals link perceived traits to mentorship conditional on gender.

Our extended model thus provides an explanation and quantifies the source of beliefs about the gender gaps in mentorship.

APPENDIX F: APPENDIX: ALTERNATE MODELS

In this section, we illustrate some examples by relaxing the linearly separable model to include interactions. These models are identified as shown in the Appendix above.

Each individual $i \in \{1, \dots, N\}$ has preferences on attributes X_j in job $j \in \{1, \dots, J\}$ given by the function

$$U_{ij} = u_i(X_j) + \epsilon_{ij}$$

In the complete scenarios individuals observe $X_j \equiv \{G_j, W_j, H_j, R_j\}$ and in the incomplete scenarios individuals observe $\widetilde{X}_j \equiv X_j \setminus \{R_j\}$. We specify the belief function of individual i as $\mathbb{E}_i(R_j \mid \widetilde{X}_j) = \widetilde{X}_j' \alpha_i$

Throughout this section, we maintain the same assumptions parallel to our instructions:

Assumption (1): All attributes not mentioned in the survey are the same for all jobs.

Assumption (2): The reported compensating differential increases only wages and changes nothing else about the job.

F.1 Model with an interaction of manager gender (G) and manager mentorship rating (R)

In this example, the utility of individual i is given by

$$U_{ij} = \sum_{x \in X} \beta_i^x x_j + \beta_i^{GR} G_j R_j + \epsilon_{ij}$$

Thus, the parameter space now contains 5 preference parameters:

$$\beta_i \equiv \{\beta_i^G, \beta_i^H, \beta_i^W, \beta_i^R, \beta_i^{GR}\}$$

and 4 belief parameters:

$$\alpha_i \equiv \{\alpha_i^G, \alpha_i^H, \alpha_i^W\}$$

Complete scenarios

In the complete scenarios, we have for all i

$$\mathbb{E}_i[U_{ij} \mid X_j] = \sum_{x \in X} \beta_i^x x_j + \beta_i^{GR} G_j R_j + \mathbb{E}_i(\epsilon_{ij} \mid X_j)$$

In any other job k that is not chosen, supposing that the individual reports Δ_{ijk} as the compensating differential, we have

$$\mathbb{E}_i[U_{ik} \mid X_k, \Delta_{ijk}] = \sum_{x \in X} \beta_i^x x_k + \beta_i^{GR} G_k R_k + \beta_i^W \Delta_{ijk} + \mathbb{E}_i(\epsilon_{ik} \mid X_k, \Delta_{ijk})$$

Given assumption (2), we have $\mathbb{E}_i(\epsilon_{ik} \mid X_k, \Delta_{ijk}) = \mathbb{E}_i(\epsilon_{ik} \mid X_k)$, which by assumption (1) is equal to $\mathbb{E}_i(\epsilon_{ij} \mid X_j)$.

Thus, normalizing $\beta_i^W = 1$, we have,

$$\Delta_{ijk} = \sum_{x \in X} \beta_i^x (x_j - x_k) + \beta_i^{GR} (G_j R_j - G_k R_k) \quad (\text{F.1})$$

Incomplete scenarios

In the incomplete scenarios, we have

$$\mathbb{E}_i[U_{ij} | \tilde{X}_j] = \sum_{x \in \tilde{X}} \beta_i^x x_j + \mathbb{E}_i(\beta_i^R R_j + \beta_i^{GR} G_j R_j + \epsilon_{ij} | \tilde{X}_j)$$

Assuming as before that individuals know their preference parameters, this simplifies to

$$\mathbb{E}_i[U_{ij} | \tilde{X}_j] = \sum_{x \in \tilde{X}} \beta_i^x x_j + \beta_i^R \mathbb{E}_i(R_j | \tilde{X}_j) + \beta_i^{GR} G_j \mathbb{E}_i(R_j | \tilde{X}_j) + \mathbb{E}_i(\epsilon_{ij} | \tilde{X}_j)$$

Using the belief function specified above, we have,

$$\mathbb{E}_i[U_{ij} | \tilde{X}_j] = \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) x_j + \beta_i^{GR} G_j \sum_{x \in \tilde{X}} \alpha_i^x x_j + \mathbb{E}_i(\epsilon_{ij} | \tilde{X}_j)$$

In any other job k that is not chosen, supposing that the individual reports $\tilde{\Delta}_{ijk}$ as the compensating differential, we have

$$\begin{aligned} \mathbb{E}_i[U_{ik} | \tilde{X}_k, \tilde{\Delta}_{ijk}] &= \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) x_k + \beta_i^{GR} G_k \sum_{x \in \tilde{X}} \alpha_i^x x_k + \beta_i^W \tilde{\Delta}_{ijk} \\ &\quad + \mathbb{E}_i(\epsilon_{ik} | \tilde{X}_k, \tilde{\Delta}_{ijk}) \end{aligned}$$

Given assumption (2), we have $\mathbb{E}_i(\epsilon_{ik} | \tilde{X}_k, \tilde{\Delta}_{ijk}) = \mathbb{E}_i(\epsilon_{ik} | \tilde{X}_k)$, which by assumption (1) is equal to $\mathbb{E}_i(\epsilon_{ij} | \tilde{X}_j)$.

Normalizing $\beta_i^W = 1$, we have

$$\tilde{\Delta}_{ijk} = \sum_{x \in \tilde{X}} \left(\beta_i^x + \beta_i^R \alpha_i^x \right) (x_j - x_k) + \beta_i^{GR} \left(G_j \sum_{x \in \tilde{X}} \alpha_i^x x_j - G_k \sum_{x \in \tilde{X}} \alpha_i^x x_k \right)$$

which simplifies to

$$\tilde{\Delta}_{ijk} = \sum_{x \in \tilde{X}} \left(\beta_i^x + \beta_i^R \alpha_i^x \right) (x_j - x_k) + \beta_i^{GR} \sum_{x \in \tilde{X}} \alpha_i^x (x_j G_j - x_k G_k) \quad (\text{E2})$$

An important difference to note in this model relative to the linearly separable model is that β_i^G alone no longer captures how much worker i values a male manager over a female manager. In this case, it is

$$U_{ij} |_{G_j=1} - U_{ij} |_{G_j=0} = \beta_i^G + \beta_i^{GR} R_j$$

Thus, the average valuation of a male manager by worker i is $\beta_i^G + \beta_i^{GR} \mathbb{E}_i(R_j)$.⁵

⁵The expectation does not condition on manager gender because the attributes are exogenously provided to the respondents and thus the mentorship rating does not significantly differ between male and female managers.

The valuation of beliefs is also different from that in the linearly separable model. This is because it is no longer weighted only by how much individuals care about mentorship quality (β_i^R) and now is augmented by how much individuals care about mentorship quality by manager gender. Using the equations above, we can show that the valuation of beliefs on mentorship for male managers relative to its counterpart for female managers in this model is

$$\alpha_i^G(\beta_i^R + \beta_i^{GR})$$

The incomplete scenarios identify $\beta_i^G + \alpha_i^G(\beta_i^R + \beta_i^{GR})$, $\beta_i^H + \beta_i^R\alpha_i^H$, $\beta_i^W + \beta_i^R\alpha_i^W$ and $\{\alpha_i^x\beta_i^{GR}\}_{x \in \{H, W\}}$. Given that the complete scenarios identify $\{\beta_i^x\}_{x \in X}$ and β_i^{GR} , the incomplete scenarios identify $\{\alpha_i^x\}_{x \in \tilde{X}}$.⁶

F.2 Model with interaction of (G) and (H) in the beliefs for rating

We specify the belief function now as

$$\mathbb{E}_i(R_j | \tilde{X}_j) = \sum_{x \in \tilde{X}} \alpha_i^x x_j + \alpha_i^{GW} G_j H_j$$

The utility function is unchanged at

$$U_{ij} = \sum_{x \in X} \beta_i^x x_j + \epsilon_{ij}$$

From the complete scenarios, following similar steps, we can derive

$$\Delta_{ijk} = \sum_{x \in X} \beta_i^x (x_j - x_k) \quad (\text{F3})$$

From the incomplete scenarios, following similar steps, we can derive

$$\tilde{\Delta}_{ijk} = \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) (x_j - x_k) + \beta_i^R \alpha_i^{GW} (G_j H_j - G_k H_k) \quad (\text{F4})$$

Following similar arguments, we can show that all parameters in this model are identified.

In this model, the valuation of a male manager relative to that of a female manager by worker i is given as β_i^G . However, the valuation of beliefs will no longer be $\beta_i^R \alpha_i^G$. The

⁶Similarly, if one were to use an interaction of the mentorship rating (R) with flexible hours (H), the corresponding complete scenario equation would be

$$\Delta_{ijk} = \sum_{x \in X} \beta_i^x (x_j - x_k) + \beta_i^{HR} (H_j R_j - H_k R_k)$$

, and the incomplete scenario equation would be

$$\tilde{\Delta}_{ijk} = \sum_{x \in \tilde{X}} (\beta_i^x + \beta_i^R \alpha_i^x) (x_j - x_k) + \sum_{x \in \tilde{X}} \alpha_i^x \beta_i^{HR} (x_j H_j - x_k H_k)$$

valuation of beliefs of worker i on male managers' mentorship ability relative to female managers' mentorship ability in job j is

$$\begin{aligned} \beta_i^R [\mathbb{E}_i(R_j | G_j = 1, W_j, H_j) - \mathbb{E}_i(R_j | G_j = 0, W_j, H_j)] \\ = \beta_i^R [\alpha_i^G + \alpha_i^{GW} H_j] \end{aligned}$$

The valuation of the corresponding beliefs by worker i is $\beta_i^R [\alpha_i^G + \alpha_i^{GW} \mathbb{E}(H_j)]$. Note that here the expectation does not vary by individuals because they observe H_j and it serves as an observable average that individuals take over jobs. Given the distribution of H_j and the identified preference and belief parameters for each worker i , the above equation identifies the distribution of the valuation of beliefs on male managers' mentorship rating relative to female managers'.

F.3 Model with $\log(\text{wages})$

In this model, the only minor difference is in the estimating equations since now the compensating differential is not separable from the wages due to the nonlinear logarithmic function. We keep other parts of the model linearly separable for simplicity; however, they can be relaxed as shown in the previous subsections.

$$U_{ij} = \beta_i^G G_j + \beta_i^H H_j + \beta_i^W \log(w_j) + \beta_i^R R_j + \epsilon_{ij}$$

Thus, the parameter space now contains 4 preference parameters:

$$\beta_i \equiv \{\beta_i^G, \beta_i^H, \beta_i^R\}$$

with β_i^W normalized to 1, and 4 belief parameters:

$$\alpha_i \equiv \{\alpha_i^G, \alpha_i^H, \alpha_i^W\}$$

Complete scenarios

In the complete scenarios, we have for all i

$$\mathbb{E}_i[U_{ij} | X_j] = \beta_i^G G_j + \beta_i^H H_j + \beta_i^W \log(w_j) + \beta_i^R R_j + \mathbb{E}_i(\epsilon_{ij} | X_j)$$

In any other job k that is not chosen, supposing that the individual reports Δ_{ijk} as the compensating differential, we have

$$\mathbb{E}_i[U_{ik} | X_k, \Delta_{ijk}] = \beta_i^G G_k + \beta_i^H H_k + \beta_i^W \log(w_k + \Delta_{ijk}) + \beta_i^R R_k + \mathbb{E}_i(\epsilon_{ik} | X_k, \Delta_{ijk})$$

Given assumption (2), we have $\mathbb{E}_i(\epsilon_{ik} | X_k, \Delta_{ijk}) = \mathbb{E}_i(\epsilon_{ik} | X_k)$, which by assumption (1) is equal to $\mathbb{E}_i(\epsilon_{ij} | X_j)$.

Thus, normalizing $\beta_i^W = 1$, we have

$$\log\left(\frac{w_k + \Delta_{ijk}}{w_j}\right) = \beta_i^G (G_j - G_k) + \beta_i^H (H_j - H_k) + \beta_i^R (R_j - R_k) \quad (\text{E5})$$

This identifies $\{\beta_i^G, \beta_i^H, \beta_i^R\}$ for each individual i . In this model, the valuation of preferences to work for a male manager relative to that of a female manager by worker i is given as β_i^G .

Incomplete scenarios

In the incomplete scenarios, we have

$$\mathbb{E}_i[U_{ij} | \tilde{X}_j] = \beta_i^G G_j + \beta_i^H H_j + \beta_i^W \log(w_j) + \beta^R \mathbb{E}_i(R_j | \tilde{X}_j) + \mathbb{E}_i(\epsilon_{ij} | \tilde{X}_j)$$

For job k and compensating differential $\tilde{\Delta}_{ijk}$, we have

$$\begin{aligned} \mathbb{E}_i[U_{ik} | \tilde{X}_k, \tilde{\Delta}_{ijk}] &= \beta_i^G G_k + \beta_i^H H_k + \beta_i^W \log(w_k + \tilde{\Delta}_{ijk}) \\ &\quad + \beta^R \mathbb{E}_i(R_k | \tilde{X}_k, \tilde{\Delta}_{ijk}) + \mathbb{E}_i(\epsilon_{ik} | \tilde{X}_k, \tilde{\Delta}_{ijk}) \end{aligned}$$

or,

$$\mathbb{E}_i[U_{ik} | \tilde{X}_k, \tilde{\Delta}_{ijk}] = \beta_i^G G_k + \beta_i^H H_k + \beta_i^W \log(w_k + \tilde{\Delta}_{ijk}) + \beta^R \mathbb{E}_i(R_k | \tilde{X}_k) + \mathbb{E}_i(\epsilon_{ik} | \tilde{X}_k)$$

This stems from the assumption that the compensating differential does not change anything about the job except for the wages. Hence, this will not change the expected rating. That is,

$$\begin{aligned} \mathbb{E}(R_k | \tilde{X}_k, \tilde{\Delta}_{ijk}) &= \mathbb{E}(R_k | \tilde{X}_k) \\ &= \alpha_i^G G_k + \alpha_i^H H_k + \alpha_i^W \log(w_k) \end{aligned}$$

Simplifying expected utility of individual i for job k and compensating differential $\tilde{\Delta}_{ijk}$ with respect to job j , we have

$$\begin{aligned} \mathbb{E}_i[U_{ik} | \tilde{X}_k, \tilde{\Delta}_{ijk}] &= (\beta_i^G + \beta_i^R \alpha_i^G) G_k + (\beta_i^H + \beta_i^R \alpha_i^H) H_k + \beta_i^W \log(w_k + \tilde{\Delta}_{ijk}) \\ &\quad + \beta_i^R \alpha_i^W \log(w_k) + \mathbb{E}_i(\epsilon_{ij} | \tilde{X}_j) \end{aligned}$$

Using arguments as above, and normalizing $\beta_i^W = 1$, we have

$$\begin{aligned} \log\left(\frac{w_k + \tilde{\Delta}_{ijk}}{w_j}\right) &= (\beta_i^G + \beta_i^R \alpha_i^G)(G_j - G_k) \\ &\quad + (\beta_i^H + \beta_i^R \alpha_i^H)(H_j - H_k) \\ &\quad + \beta_i^R \alpha_i^W (\log(w_j) - \log(w_k)) \end{aligned} \tag{F.6}$$

This identifies $\{\beta_i^G + \beta_i^R \alpha_i^G, \beta_i^H + \beta_i^R \alpha_i^H, \beta_i^R \alpha_i^W\}$ for each individual i . Thus valuation of beliefs on male manager mentorship which is given as $\beta_i^R \alpha_i^G$ is identified from the variation between the complete and the incomplete scenarios.

F.4 Model with scaled mentorship rating $f(R_j)$

In this sub-section we explore identification when we allow utility over mentorship, R to be linear in $\beta_i^R f(R_j)$ for a monotone increasing $f(\cdot)$ instead of being linear in $\beta_i^R R_j$.

Indeed, since individuals hold $f(\cdot)$ fixed while reporting their choices and compensating differentials, our exogenous variations in the survey cannot non-parametrically separately identify $f(\cdot)$ from β_i^R . This is true even if $f(\cdot)$ does not vary by individuals. However, for any given $f(\cdot)$, the preference and belief parameters are non-parametrically identified.

To see identification, define $X_j \equiv \{G_j, W_j, H_j, f(R_j)\}$. Each individual $i \in \{1, \dots, N\}$ has preferences over attributes X_j in job $j \in \{1, \dots, J\}$ given by

$$U_{ij} = \sum_{x \in X} \beta_i^x x_j + \epsilon_{ij}$$

In the incomplete scenarios, where individuals do not observe R_j , we now specify the belief function of individual i as $\mathbb{E}_i [f(R_j) \mid \widetilde{X}_j] = \widetilde{X}_j' \alpha_i$ where $\widetilde{X}_j \equiv X_j \setminus f(R_j)$. Given this the proof of identification of the parameters follows identical steps to the baseline case discussed in the main text and shown in Appendix A.

Hence, the intuition of identification remaining the same as before, the additional modification which facilitates identification in this set-up is to specify the belief function as $\mathbb{E}_i [f(R_j) \mid \widetilde{X}_j]$ for any given $f(\cdot)$ instead of $\mathbb{E}_i [R_j \mid \widetilde{X}_j]$. Additionally, note that in allowing preferences over mentorship to be linear in $\beta_i^R f(R_j)$ for a monotone increasing $f(\cdot)$, even though the parameters of the model are identified, the interpretation of the parameters change. In particular, the preference parameter β_i^R tells us the willingness to forgo wages for a unit increase in $f(R_j)$ instead of a unit increase in R_j . This interpretation is parallel to the argument of non-identification of $f(\cdot)$. If $f(\cdot)$ were identified, we could have identified the willingness to forgo wages for a unit increase in R_j . To see this observe that, if $f(\cdot)$ were identified, then $\frac{\partial f(R_j)}{\partial R_j}$ is identified, and hence the willingness to forgo wages for a unit increase in R_j i.e., $\beta_i^R \frac{\partial f(R_j)}{\partial R_j}$ would have been identified.

F.5 Model with mentorship quality as a proxy for overall manager quality

In this section, we delineate a more generic model than the one presented in the main paper. The identifying assumptions remain the same. However, we relax the interpretation of the attribute of manager mentorship ability. In this specification, individuals care about overall manager quality (Q) in addition to caring about wages, flexibility of hours and manager gender. The mentorship rating acts as a signal of overall manager quality. Individuals care about this overall manager quality. The purpose of this generic model is to show that the finding of belief-based discrimination still holds.

Redefining the set of attributes to $A \equiv \{G, W, H, Q\}$, the utility of individual i takes the same linear form:

$$U_{ij} = \sum_{x \in A} \beta_i^x x_j + \epsilon_{ij} \quad (\text{F.7})$$

Observe that now in both the complete and incomplete scenarios, individuals need to form expectations on manager quality. In the incomplete scenarios, they do not have information on the manager's mentorship rating, whereas in the complete scenarios, they do. The expected utilities in the complete and incomplete scenarios take the following forms:

$$\begin{aligned} \text{Incomplete Scenarios: } \mathbb{E}_i[U_{ij} | \tilde{X}_j] &= \sum_{x \in A \setminus Q} \beta_i^x x_j + \beta_i^Q \mathbb{E}_i(Q_j | \tilde{X}_j) + \mathbb{E}_i(\epsilon_{ij} | \tilde{X}_j) \\ \text{Complete Scenarios: } \mathbb{E}_i[U_{ij} | X_j] &= \sum_{x \in A \setminus Q} \beta_i^x x_j + \beta_i^Q \mathbb{E}_i(Q_j | X_j) + \mathbb{E}_i(\epsilon_{ij} | X_j) \end{aligned} \quad (\text{F.8})$$

As explained above, in both expected utilities, the individuals forms expectations on manager quality. However, in the complete scenario, the individual has the additional information of the manager's mentorship rating. We parameterize the expectation on manager quality in the following way:

$$\begin{aligned} \text{Complete Scenarios: } \mathbb{E}_i(Q_j | X_j) &= \sum_{x \in A \setminus Q} \gamma_i^x x_j + \gamma_i^R R_j \\ \text{Incomplete Scenarios: } \mathbb{E}_i(Q_j | \tilde{X}_j) &= \sum_{x \in A \setminus Q} \gamma_i^x x_j + \gamma_i^R \mathbb{E}[R_j | \tilde{X}_j] \\ &= \sum_{x \in A \setminus Q} \tilde{\gamma}_i^x x_j \end{aligned} \quad (\text{F.9})$$

Incorporating the above in the expected utility functions in both scenarios, we have

$$\begin{aligned} \text{Complete Scenarios: } \mathbb{E}_i[U_{ij} | X_j] &= \sum_{x \in A \setminus Q} (\beta_i^x + \beta_i^Q \gamma_i^x) x_j + \beta_i^Q \gamma_i^R R_j + \mathbb{E}_i(\epsilon_{ij} | X_j) \\ \text{Incomplete Scenarios: } \mathbb{E}_i[U_{ij} | \tilde{X}_j] &= \sum_{x \in A \setminus Q} (\beta_i^x + \beta_i^Q \tilde{\gamma}_i^x) x_j + \mathbb{E}_i(\epsilon_{ij} | \tilde{X}_j) \end{aligned} \quad (\text{F.10})$$

Then, with the same set of identifying assumptions, given the reported compensating differentials and normalizing $\beta^W = 1$, we have the following indifference conditions:

$$\begin{aligned}\Delta_{ijk} &= \sum_{x \in A \setminus Q} \underbrace{(\beta_i^x + \beta_i^Q \gamma_i^x)}_{\beta_i^{x(CS)}} (x_j - x_k) + \beta_i^Q \gamma_i^R (R_j - R_k) \\ \tilde{\Delta}_{ijk} &= \sum_{x \in A \setminus Q} \underbrace{(\beta_i^x + \beta_i^Q \tilde{\gamma}_i^x)}_{\beta_i^{x(IS)}} (x_j - x_k)\end{aligned}\tag{F11}$$

The differences in the coefficients in front of the gender differences across the complete and incomplete scenarios give us

$$\beta_i^{G(CS)} - \beta_i^{G(IS)} = \beta_i^Q (\gamma_i^G - \tilde{\gamma}_i^G)\tag{F12}$$

γ encapsulates the information on manager quality given X_j , whereas $\tilde{\gamma}$ encapsulates the information on manager quality given $\tilde{X}_j$, i.e., in the absence of the information on manager quality. In the presence of belief-based discrimination against female managers, this should be negative. This is what our estimates show, given that individuals care positively about manager quality. Thus, under this model specification, belief-based discrimination is identified. Observe the analogy with the model presented in the main paper. Here, too, if the individual does not care about manager quality (i.e., $\beta^Q = 0$), the parameters identified from the complete and incomplete scenarios must be identical because the variation in information revelation will have no effect.

APPENDIX G: APPENDIX FIGURES AND TABLES

FIGURE G.1. Survey definitions and instructions
(a) Definitions of attributes

INSTRUCTIONS

In each question you will see a choice scenario.

A scenario will have 3 jobs (X, Y and Z), which you have to assume have been offered to you.

Each job will have 4 characteristics:

Manager: First name of the team's manager.

Annual wages: Gross annual salary (in lakhs).

Flexible hours: Whether the job allows for flexible hours or not.

Manager rating: This is the average rating of the mentorship of the manger, provided by this manager's current employees in an anonymous survey.

This is a measure of how good of a mentor is this manager to its subordinates.

This rating is on a scale of 1-5. The scale points mean as follows: **1: Poor; 2: Fair 3: Good, 4: Very good and 5: Excellent.**

There will be 20 such choice scenarios.

(b) Instructions

INSTRUCTIONS

In each scenario we will first ask you:

A. To choose one job among 3 job options.

Your instructions are:

To assume that all other characteristics, which are NOT MENTIONED here, are THE SAME in all the jobs. For example work from home under each manager is not shown here. You are to assume that its either available or not available in all three jobs. No job is different in anything that you do not observe,

In each scenario, we will then ask you for:

B. If you were to negotiate wages, how much **minimum increase in wages** you would need in each of the other two jobs, for you choose them instead.

Your instructions are:

State your wage increase, assuming that it does not change anything else about the job.

Let us give you an example from a survey of food preference.

A. Here you will see me choose one dish among 3 different dishes.

B. Then you will see me state how much MINIMUM drop in prices in the other two dishes would make me choose each of them instead.

FIGURE G.1. Notes: Panel A shows the first part of the instructions that defined the job attributes used in the survey, and Panel B shows remaining instructions shown to respondents.

TABLE G.1. Incomplete scenario adapted example for representative US jobs

Job choice

Rating of each manager could be different, but the data is unavailable. Anything else that you don't see here, is the SAME across all jobs. Please select your most preferred job.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	35k	39k	37k
Flexible hours	yes	no	yes
Manager	John	Susan	Robert
Manager rating	N/A	N/A	N/A

Job X

Job Y

Job Z

Compensating differential (if job chosen was Y)

You chose Job Y.

If you were to negotiate your wage, how much of a MINIMUM INCREASE IN WAGES would you need in each of the other jobs for you to choose it instead of Job Y? The scale here ranges from 0 to 5,000 USD.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	35k	39k	37k
Flexible hours	yes	no	yes
Manager	John	Susan	Robert
Manager rating	N/A	N/A	N/A

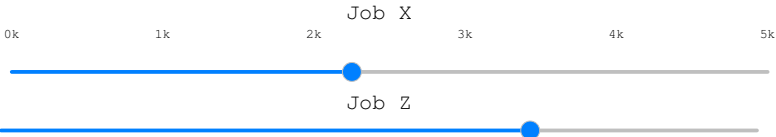


TABLE G.1. Notes: Jobs X and Z have male managers, and Job Y has a female manager. Across the 10 incomplete scenarios, five scenarios have two jobs with male managers and one job with a female manager, and the remaining five have two jobs with female managers and one job with a male manager.

TABLE G.2. Complete scenario adapted example for representative US jobs

Job choice

Anything else that you don't see here, is the SAME across all jobs. Manager rating is on a scale of 1-5. 1: Poor; 2: Fair; 3: Good; 4: Very good; 5: Excellent. Please select your most preferred job.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	40k	42k	45k
Flexible hours	yes	no	yes
Manager	James	Barbara	Mary
Manager rating	3.70	4.00	3.15

Job X

Job Y

Job Z

Compensating differential (if job chosen was X)

You chose Job X.

If you were to negotiate your wage, how much of a MINIMUM INCREASE IN WAGES would you need in each of the other jobs for you to choose it instead of Job X? The scale here ranges from 0 to 5,000 USD.

Attributes	JOB X	JOB Y	JOB Z
Annual Wages	40k	42k	45k
Flexible hours	yes	no	yes
Manager	James	Barbara	Mary
Manager rating	3.70	4.00	3.15

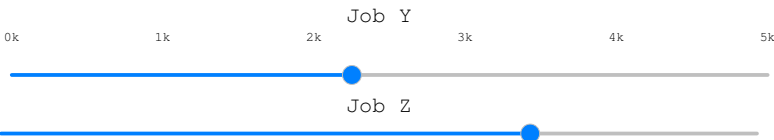


TABLE G.2. *Notes:* Jobs X and Y have male managers, and Job Z has a female manager. Overall, across the 10 complete scenarios, five scenarios have two jobs with male managers and one job with a female manager, and the remaining five have two jobs with female managers and one job with a male manager.

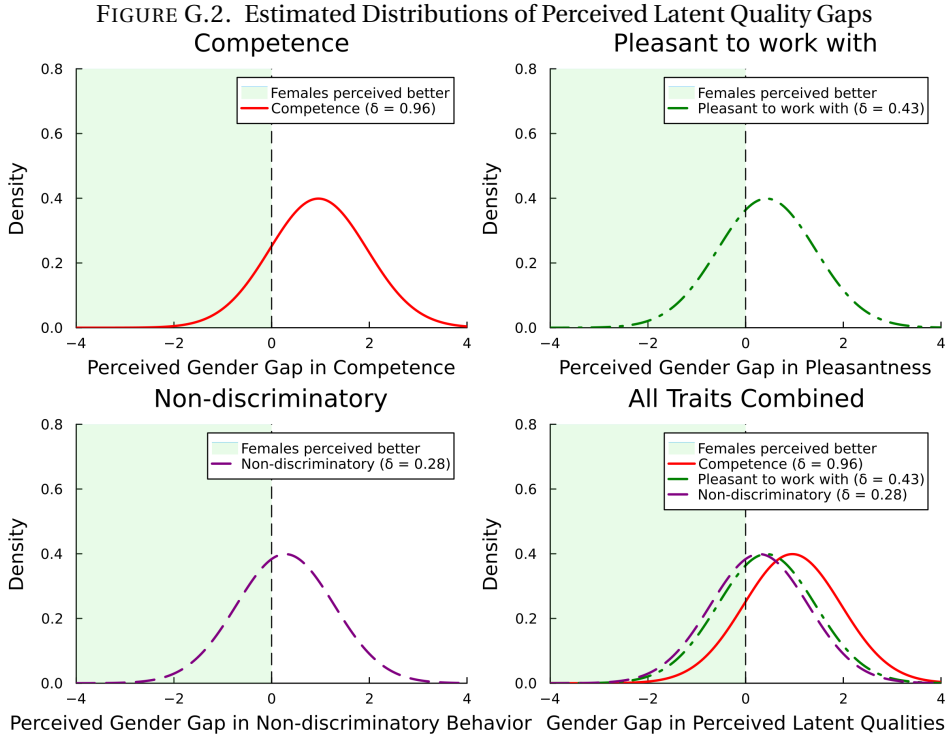


FIGURE G.2. Notes: Each panel shows the estimated distribution of perceived gender gaps in specific managerial qualities. The light green shaded region ($Q_{iq}^F > Q_{iq}^M$) represents the area where females are perceived to be strictly better than males. Distribution for trait q are estimated as $\text{Normal}(\delta_q, 1)$ where δ_q represents the mean perceived gap for trait q which in turn are estimated from the ordered probit models. The bottom-right panel overlays all three distributions for comparison.

TABLE G.3. Jointly Estimated Preference and Belief Parameter Estimates with Perception on other traits and Decomposition of beliefs

Panel A: Complete and Incomplete Scenarios:					
Incomplete Scenarios			Complete Scenarios		
Parameters	in 10 ⁵ INR	% of wages	Parameters	in 10 ⁵ INR	% of wages
$\tilde{\beta}_i^{G,adj}$	-0.006 (0.005)	0.1%	$\beta_i^{G,adj} = \beta_i^G + \sum_q \beta_i^q \delta_q$ (Male Manager)	-0.045 (0.007)	0.6%
$\tilde{\beta}_i^H$	1.134 (0.059)	16.0%	β_i^H (Flexible Hours)	0.777 (0.027)	10.9%
$\tilde{\beta}_i^W$	1.369 (0.068)	19.3%	β_i^W (Annual Wages)	1	
			β_i^R (Mentorship)	0.792 (0.030)	11.1%
Observations	11,600			11,600	
Panel B: Trait-Specific Components and δ Values					
Parameters	Estimate	δ_q	Correlation b/w trait q and mentorship	Estimate	
β_i^{q1}	-0.009 (0.007)	0.958	$\beta_i^R \alpha_i^{q1}$	0.015 (0.008)	
β_i^{q2}	0.002 (0.004)	0.280	$\beta_i^R \alpha_i^{q2}$	0.004 (0.005)	
β_i^{q3}	-0.001 (0.003)	0.428	$\beta_i^R \alpha_i^{q3}$	0.004 (0.004)	
Panel C: Valuation of Belief Parameters and Decomposition					
Parameter	Estimate	% of wages	Decomposition		
$\beta_i^R (\alpha_i^G + \sum_q \alpha_i^q \delta_q)$	0.039 (0.009)	0.6%	Trait Driven:	0.017 (0.009)	
			Residual:	0.022 (0.008)	
$\beta_i^R \alpha_i^H$	0.358 (0.044)	5.0%			
$\beta_i^R \alpha_i^W$	0.369 (0.068)	5.2%			

Notes: Panel A shows estimates from the jointly estimated model with trait adjustments. The complete scenario estimate for male managers includes trait-specific adjustments: $\beta_i^{G,adj} = \beta_i^G + \sum_{q=1}^3 \beta_i^q \delta_q$. The incomplete scenario parameter is $\tilde{\beta}_i^{G,adj} = \beta_i^G + \sum_q \beta_i^q \delta_q + \beta_i^R (\alpha_i^G + \sum_q \alpha_i^q \delta_q) + \sum_q (\beta_i^q + \beta_i^R \alpha_i^q) \eta_{iq}$. Panel B displays the trait-specific components where δ_q represents the average difference in trait q between male and female managers. Quality 1 (q_1) is competence, quality 2 (q_2) is non-discriminatory, and quality 3 (q_3) is pleasant to work with. Panel C shows belief parameters where the total valuation $\beta_i^R (\alpha_i^G + \sum_q \alpha_i^q \delta_q)$ is decomposed into trait-driven effects $\beta_i^R \sum_q \alpha_i^q \delta_q$ and residual effects $\beta_i^R \alpha_i^G$, with trait-driven effects explaining 43.6% of the gender gap. Standard errors are computed using block bootstrap at the individual level with 1,000 replications.

FIGURE G.3. Expected rating

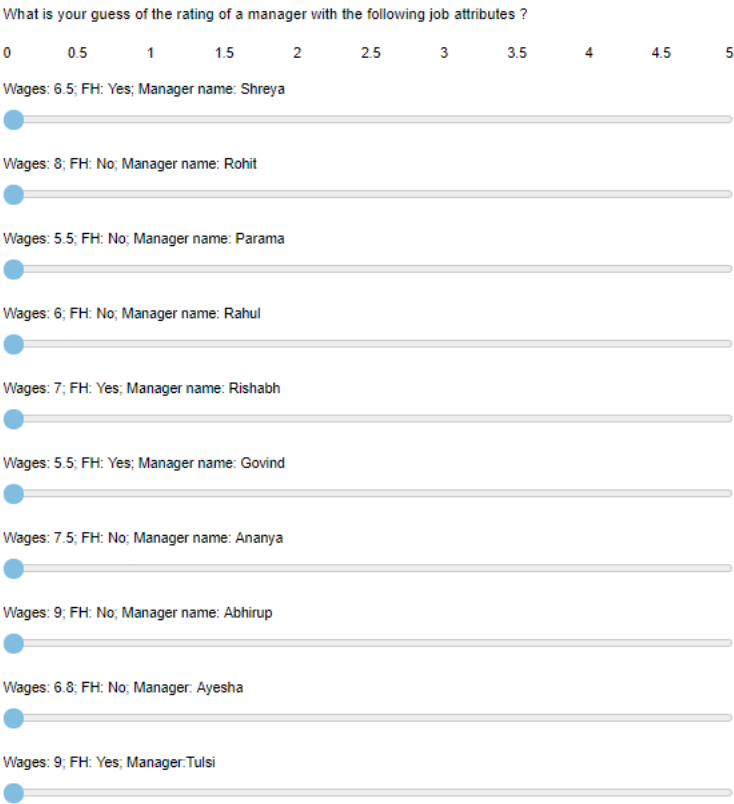


TABLE G.4. Estimates of expected beliefs from directly elicited belief data

	All	Female	Male
Directly Elicited Beliefs			
Male Manager (α^G)	-0.042 (0.012)	-0.027 (0.016)	-0.053 (0.016)
Flexible Hours (α^H)	0.321 (0.023)	0.321 (0.033)	0.321 (0.031)
Annual Wages (α^W)	0.278 (0.013)	0.290 (0.018)	0.270 (0.018)
Individual Fixed Effects	Yes	Yes	Yes
N	5,800	2,420	3,380
R^2	0.598	0.609	0.592

TABLE G.4. *Notes:* Standard errors are clustered at the individual level. These estimates are obtained from a fixed effects regression of directly elicited manager rating on the exogenously varied job attributes: indicator for male manager, flexibility of hours, and annual wages. Specifically, we estimate $R_{ij}^{direct} = \theta_i + \alpha^G G_j + \alpha^H H_j + \alpha^W W_j + \eta_{ij}$, where R_{ij}^{direct} is the directly elicited rating of manager in job j by individual i , G_j is an indicator of male manager, H_j is an indicator of flexible hours, and W_j is the annual wages of job j , and θ_i is an individual fixed effect.